HPSC FSO Master Handout — Food Toxicology, Inherent Toxicants, Contaminants, Residues & Heavy Metals

Read First: Method & Sources

HPSC Food Safety Officer (Group-B) — Master Study Handout

Module: Food Toxicology, Inherent Plant and Animal Toxicants, Contaminants, Residues and Heavy Metals

This handout is written as a Senior Professor and Chief Paper Setter would teach: molecule first, then clinical pathology, then analytical identification, then the statutory ceiling, then the answer skeleton. Every section is complete; nothing is left as a placeholder.

1. Source hierarchy used in this handout

Rank	Authority	Use in this handout
1	Food Safety and Standards Act, 2006 and the Food Safety and Standards (Contaminants, Toxins and Residues) Regulations, 2011, as amended; FSS (Prohibition and Restrictions on Sales) Regulations, 2011; FSS (Food Products Standards and Food Additives) Regulations, 2011	Statutory ceilings, prohibitions, zero-tolerance lists
2	Codex Alimentarius, JECFA and JMPR evaluations	Acceptable Daily Intake, Maximum Residue Limit methodology, international benchmarks where Indian text is silent
3	Reference texts: Parmar, Jain and Aparnathi, Food Chemistry (Lesson 21 on toxic trace elements and antinutritional factors); Srilakshmi, Food Science; FSN 421 notes on hazards, additives and plant toxicants	Mechanistic and compositional teaching material

Statutory numerical limits change by gazette amendment. The limits printed in this handout are representative values from the 2011 Regulations and their 2018 to 2020 harmonisation amendments. Before the examination, re-verify the exact figure for any commodity against the latest notification. A question that says 'according to the regulation' is asking for the statutory number, not a textbook number.

2. Examination architecture (HPSC pattern)

Stage 1: Objective Screening Test

- 100 marks, multiple-choice, penalty of 0.25 mark per wrong answer.
- Favourite formats: count-based pair questions (Only 1 / Only 2 / All 3 / None), assertion-reason, match-the-following, and absolute-qualifier statements using words such as always, only, never, all, completely.
- Expected score of answering a question with probability p of being right: E = 1.25p minus 0.25. Answering is mathematically favourable when p is greater than 0.20, that is, when you can eliminate at least one of four options with confidence.

Stage 2: Subject Knowledge Test

- 150 marks, 3 hours, conventional descriptive answers, carrying 87.5 percent weightage in final merit.
- Time budget: 1.2 minutes per mark. A 10-mark answer gets about 12 minutes; a 15-mark answer about 18 minutes. Reserve 10 to 15 minutes for planning and review.
- Scoring architecture for a toxicology answer: Definition, Source and chemistry, Mechanism (with a labelled pathway), Clinical pathology, Analytical identification, Statutory position, Processing or control, Conclusion.

Stage 1 method for count-based pairs

1. Treat each pair as an independent True or False item.
2. Underline the subject, the predicate and the qualifier.
3. Assign c = 1 for a correct pair and c = 0 for a wrong pair.
4. Compute N = c1 + c2 + c3 and map N to the option.
5. Never let the familiarity of the topic decide the answer; the predicate decides.

3. Absolute-qualifier checklist

Statement as set in a paper	Why it is a trap	Correct reading
Cooking always destroys plant toxicants	Glycoalkaloids, oxalates (insoluble), BOAA in dry heat and phytate survive ordinary cooking	Detoxification is toxin-specific
All shellfish toxins come from dinoflagellates	Domoic acid comes from diatoms (Pseudo-nitzschia)	Three syndromes are dinoflagellate-linked; ASP is diatom-linked
All sodium-channel toxins are blockers	Brevetoxin and ciguatoxin are activators (site 5)	Saxitoxin and tetrodotoxin block site 1; brevetoxin and ciguatoxin open the channel
Scombroid poisoning is a fish allergy	It is histamine intoxication mimicking allergy (pseudo-allergic)	Not immunoglobulin E mediated; occurs in non-allergic people
Methylmercury is removed by cooking	Organomercury is stable and protein-bound	Only source control reduces exposure
Carbamate poisoning is treated like organophosphate poisoning	Carbamylation is reversible; pralidoxime is not routinely indicated	Atropine for both; oxime mainly for organophosphates
Chloramphenicol has an MRL	It is on the zero-tolerance (prohibited) list	No residue is permitted
ADI equals NOAEL multiplied by 100	ADI is NOAEL divided by the safety factor	ADI = NOAEL / 100 typically

4. One-page numerical anchor sheet

Item	Number to memorise
Potato glycoalkaloids (solanine plus chaconine)	Safe ceiling 200 mg/kg fresh tuber (20 mg/100 g); toxic at 2 to 5 mg/kg body weight
Hydrogen cyanide lethal dose	About 0.5 to 3.5 mg/kg body weight; Codex ceilings 50 mg/kg for sweet cassava and 10 mg/kg for cassava flour (as HCN)

Item	Number to memorise
Nitric acid test for argemone oil	Detects about 0.25 percent and above; paper chromatography detects about 0.0001 percent
Paralytic shellfish poisoning action level	80 micrograms saxitoxin equivalents per 100 g (0.8 mg/kg)
Amnesic shellfish poisoning action level	20 mg domoic acid per kg (20 micrograms per gram)
Diarrhetic shellfish poisoning (European Union)	160 micrograms okadaic acid equivalents per kg
Histamine in fish	Codex: 100 mg/kg hygiene indicator; 200 mg/kg decomposition limit; United States action level 50 mg/kg
Tin in canned foods	250 mg/kg (canned foods); 150 mg/kg (canned beverages)
Mercury in fish	0.5 mg/kg; 1.0 mg/kg methylmercury in large predatory fish
Inorganic arsenic in polished rice	0.2 mg/kg (Codex and FSSAI harmonised value); 0.35 mg/kg husked rice (Codex)
Lead in milk and infant formula	0.02 mg/kg (harmonised value)
Default pesticide MRL where none is fixed	0.01 mg/kg (FSSAI, 2018 amendment)
Safety factor for ADI derivation	100 (10 interspecies multiplied by 10 intraspecies)
Acrylamide formation threshold	Above about 120 degrees Celsius in low-moisture asparagine-rich starchy food

5. How to use the tabs

- Tabs 2 and 3: Part 1, inherent plant toxicants and antinutritional factors.
- Tab 4: Part 2, marine and animal biotoxins.
- Tabs 5 and 6: Part 3, heavy metals, agrochemical and veterinary residues, process-induced contaminants.
- Tab 7: Part 4, master comparative tables, four-question count-based revision bank with arithmetic verification, and the two Stage 2 model answers.

Part 1A: Legume Toxicants

PART 1A: LEGUME TOXICANTS AND ANTINUTRITIONAL FACTORS

First-principles orientation

Legumes are seeds. A seed must survive insects, fungi and predators until germination, so it stores chemical defences: neurotoxic amino acids, cyanide-releasing glycosides, pro-oxidant glycosides, carbohydrate-binding proteins, protease inhibitors and phosphorus locked as phytate. Human processing (soaking, germination, fermentation, moist heat) is essentially a set of counter-strategies against these defences. Every toxicant below is best understood by asking three questions: what is the molecule, what does it hit in the human body, and which processing step breaks it.

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1. Lathyrism and Khesari Dal (Lathyrus sativus)

1.1 Botany and epidemiology

- Lathyrus sativus is called khesari, kesari dal, teora, chickling vetch or grass pea. It is drought-hardy and cheap, so it dominates the diet of the poorest cultivators in Madhya Pradesh, Chhattisgarh, eastern Uttar Pradesh, Bihar, Odisha and Maharashtra in drought years.
- Neurolathyrism appears when khesari supplies more than about 30 percent of dietary energy for 2 to 6 months, typically in young adult males aged 15 to 45, often during famine when it is eaten unprocessed as roasted seeds or coarse flour.
- Historical context: Sleeman (1844) described it in Central India; the Lathyrism Enquiry Field Unit of the Indian Council of Medical Research worked at Rewa in the 1950s and 1960s.

1.2 The molecule

- Toxin: beta-N-oxalyl-L-alpha,beta-diaminopropionic acid, abbreviated BOAA, also called ODAP (3-N-oxalyl-L-2,3-diaminopropanoic acid).
- Structure: a non-protein amino acid in which the beta-amino group of L-2,3-diaminopropionic acid is acylated by oxalic acid. Two isomers exist: beta-ODAP (the neurotoxic form, about 95 percent of the total) and alpha-ODAP (far less toxic); heating can interconvert them.
- Content: 0.1 to 2.5 percent of seed dry weight depending on variety and drought stress. Low-toxin lines (ODAP below 0.1 percent) such as Pusa 24 and Ratan have been bred.
- It is water-soluble, which is the entire basis of the detoxification chemistry.

1.3 Mechanism of neurotoxicity

```
BOAA / ODAP (structural analogue of glutamate)
        |
        v
Agonist at AMPA-type ionotropic glutamate receptors on motor neurons
        |
        v
Sustained sodium and calcium influx (excitotoxicity)
        |
        v
Mitochondrial complex I inhibition, reactive oxygen species,
glutathione depletion, inhibition of glutamate uptake by astrocytes
        |
```

v
Selective degeneration of Betz cells and the lateral corticospinal (pyramidal) tracts supplying the lower limbs
|
v
Irreversible spastic paraparesis

- Why lower limbs: the longest corticospinal axons, those to the legs, are most vulnerable to energy failure.
- Synergistic risk factors: low intake of sulphur amino acids (methionine and cysteine deplete glutathione), zinc deficiency, hard physical labour, and febrile illness.

1.4 Clinical pathology

Stage	Presentation
Prodrome	Calf cramps, heaviness of legs, nocturnal spasms after a heavy khesari meal
Latent (no-stick) stage	Spastic gait but walks unaided
One-stick stage	Needs one stick; scissoring gait, crossed adductor spasm, toe-walking
Two-stick stage	Needs two sticks or crutches
Crawler stage	Cannot stand; moves on all fours

- Upper motor neuron signs: hypertonia, exaggerated knee and ankle jerks, ankle clonus, extensor plantar response (Babinski sign). Sensation, bladder and intellect are usually spared.
- The lesion is irreversible; there is no antidote. Prevention is entirely dietary and regulatory.

1.5 Processing detoxification (all exploit water solubility)

Method	Procedure	Removal of ODAP
Steeping in hot water	Soak dehusked dal in 2 to 3 volumes of water at 60 to 70 degrees Celsius for 1 hour, discard water, wash	70 to 80 percent
Parboiling	Soak overnight, steam or boil in excess water for 15 to 20 minutes, discard water, sun-dry	80 to 90 percent
Lime soaking	Soak in 1 to 2 percent calcium hydroxide (lime water) for several hours, wash; alkaline pH hydrolyses the oxalyl amide bond	Up to 90 percent
Fermentation	Lactic fermentation (as in idli or dhokla batter)	Partial, variable
Dry roasting or frying	Little or no removal; dry heat does not remove a water-soluble toxin	Negligible

The examiner's favourite pairing: the method must be a wet method with discarding of the soak water. Any question stating that roasting detoxifies khesari is false.

1.6 Statutory position

- The Prevention of Food Adulteration Rules (Rule 44A, 1961) prohibited the sale of khesari gram and its products in any form.
- This continues under the Food Safety and Standards (Prohibition and Restrictions on Sales) Regulations, 2011, which prohibit the sale of kesari gram (Lathyrus sativus), kesari dal, kesari flour and any admixture of kesari with other grams, dal or flour.

- In 2016 the FSSAI scientific panel reviewed lifting the ban in view of low-toxin varieties, but the sale prohibition remains the examinable legal position; a notification lifting it has not been finalised.
- Analytical detection in adulterated dal: khesari dal is wedge-shaped with a square truncated end, unlike the rounder lentil or red gram; the chemical test is the 50 percent hydrochloric acid test (pink colour on heating with the dal in dilute hydrochloric acid indicates khesari).

[HPSC EXAM TRAP] Neurolathyrism (from Lathyrus sativus, BOAA) must not be confused with osteolathyrism (from Lathyrus odoratus, sweet pea, caused by beta-aminopropionitrile, which inhibits lysyl oxidase and weakens collagen). A pair that assigns BOAA to skeletal deformity is false.

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2. Cyanogenic Glycosides

2.1 The molecules and their sources

Glycoside	Aglycone type	Sugar	Main foods	Final hydrolysis products
Amygdalin	Mandelonitrile (aromatic)	Gentiobiose (two glucose)	Bitter almonds, apricot, peach, plum and cherry kernels, apple seeds	Benzaldehyde + HCN + 2 glucose
Prunasin	Mandelonitrile	Glucose	Intermediate in amygdalin breakdown; wild cherry	Benzaldehyde + HCN
Linamarin	Acetone cyanohydrin (aliphatic)	Glucose	Cassava (tapioca), lima beans, flax	Acetone + HCN
Lotaustralin	Methyl ethyl ketone cyanohydrin	Glucose	Cassava (minor), lima beans	Butanone + HCN
Dhurrin	p-Hydroxymandelonitrile	Glucose	Sorghum (jowar), especially young shoots	p-Hydroxybenzaldehyde + HCN

- The intact glycoside is non-toxic. The plant stores it in the vacuole and keeps the hydrolysing enzyme in a separate compartment (cell wall or cytoplasm). Chewing, grating, grinding or insect damage breaks the compartments.

2.2 Enzymatic hydrolysis

```
Cyanogenic glycoside
      | beta-glucosidase (linamarase in cassava, emulsin in almonds)
      v
Sugar + cyanohydrin (alpha-hydroxynitrile)
      | hydroxynitrile lyase, or spontaneous at pH above 5 and warm temperature
      v
Carbonyl compound (benzaldehyde, acetone) + hydrogen cyanide (HCN)
```

- Gut bacteria also possess beta-glucosidases, so swallowed intact glycoside can still release cyanide in the colon. This is why 'eat it raw without chewing' is not safe.

2.3 Mechanism of cyanide toxicity

- Cyanide ion binds the ferric iron (Fe3+) of cytochrome c oxidase (complex IV) in the mitochondrial electron transport chain.
- Electron transfer to oxygen stops; oxidative phosphorylation ceases; cells switch to anaerobic glycolysis, producing lactic acidosis.
- Result is histotoxic hypoxia: arterial blood stays bright red and oxygen-saturated, but tissues cannot use it. Venous blood is also bright red, a classical sign.

- Detoxification in the body: the mitochondrial enzyme rhodanese transfers sulphur from thiosulphate to cyanide, forming the far less toxic thiocyanate, which is excreted in urine. This is why sodium thiosulphate is an antidote and why chronic cassava eaters have high urinary thiocyanate (and secondary goitre, because thiocyanate blocks iodide uptake).

2.4 Clinical pathology

Form	Picture
Acute poisoning	Headache, dizziness, bitter-almond odour, tachypnoea then respiratory depression, convulsions, coma, death within minutes to hours; lethal oral dose of HCN about 0.5 to 3.5 mg/kg body weight (50 to 200 mg in an adult)
Konzo	Abrupt, permanent, symmetrical spastic paraparesis in cassava-dependent populations of Central and East Africa during drought, linked to inadequately processed bitter cassava plus protein (sulphur amino acid) deficiency
Tropical ataxic neuropathy	Chronic sensory ataxia, optic atrophy, deafness in long-term low-dose cassava eaters (Nigeria)
Endemic goitre and cretinism	Thiocyanate competes with iodide at the thyroid sodium-iodide symporter

- Antidotes: hydroxocobalamin (binds cyanide to form cyanocobalamin), sodium nitrite (induces methaemoglobin, which scavenges cyanide), sodium thiosulphate (substrate for rhodanese), and 100 percent oxygen.

2.5 Detoxification by processing

Each step either frees the HCN (by letting the enzyme act) or removes it (HCN is volatile, boiling point 26 degrees Celsius, and water-soluble).

Step	Chemistry	Note
Peeling	Peel of bitter cassava carries most of the linamarin	First step, large reduction
Grating or crushing	Ruptures compartments so linamarase meets linamarin	Essential to liberate HCN before heating
Soaking in water (3 to 5 days, retting)	Hydrolysis plus leaching of soluble glycoside and cyanide into water	Discard the water
Sun-drying after wetting ('wetting method')	Slow drying allows enzyme hydrolysis then volatilisation of HCN	Rapid drying traps intact glycoside
Fermentation (gari, fufu, lafun)	Lactic bacteria and endogenous enzymes complete hydrolysis; acid pH keeps cyanohydrin stable so a later heating step is needed	Gari limit 2 mg/kg HCN (Codex)
Boiling in excess water	Volatilises HCN and destroys enzyme; must follow grating, because boiling first inactivates the enzyme and locks the glycoside inside	Order of operations matters

- Statutory and Codex ceilings: sweet cassava 50 mg/kg total HCN; cassava flour 10 mg/kg; gari 2 mg/kg. Bitter cassava carries more than 100 mg/kg and up to 1,000 mg/kg fresh weight.

[HPSC EXAM TRAP] Boiling intact chunks of bitter cassava without grating is less effective than grating followed by soaking, because the enzyme that liberates HCN is itself denatured by heat. Sun-drying is only effective if the tissue is first crushed or kept moist long enough for hydrolysis. Also note that amygdalin is wrongly marketed as 'vitamin B17' or laetrile; it has no vitamin status and causes cyanide poisoning.

3. Favism and Faba Beans (Vicia faba)

3.1 The molecules

- Vicine and convicine are pyrimidine beta-glucosides concentrated in the cotyledons of faba (broad) beans at 0.5 to 1.0 percent dry weight.
- Hydrolysis by beta-glucosidase (in the bean, in gut bacteria and in the intestinal mucosa) removes glucose to give the aglycones divicine (from vicine) and isouramil (from convicine).
- Divicine and isouramil are unstable 2,6-diaminopyrimidine-diols that autoxidise rapidly, generating hydrogen peroxide and superoxide radicals and oxidising reduced glutathione (GSH) to oxidised glutathione (GSSG).

3.2 The genetic lesion: G6PD deficiency

```
Glucose-6-phosphate
      | glucose-6-phosphate dehydrogenase (G6PD), first step of the pentose phosphate pathway
      v
6-Phosphogluconolactone + NADPH
      |
      v
NADPH is the only source of reducing power in the mature red cell
      |
      v
Glutathione reductase uses NADPH to regenerate GSH from GSSG
```

- The red blood cell has no mitochondria and no nucleus; the pentose phosphate pathway is its sole NADPH source.
- In G6PD deficiency (X-linked, hence mainly affecting males; common in Mediterranean, Middle Eastern, African and Indian populations, including Parsis and some tribal groups), NADPH supply fails under oxidant stress.
- Divicine and isouramil deplete GSH faster than it can be regenerated; haemoglobin is oxidised to methaemoglobin and denatured into Heinz bodies that precipitate on the membrane; the spleen removes damaged cells (extravascular haemolysis) and some cells lyse in circulation (intravascular haemolysis).

3.3 Clinical pathology

- Acute haemolytic anaemia 24 to 48 hours after eating fresh or dried faba beans (or inhaling pollen in sensitive children): pallor, jaundice, dark red-brown urine (haemoglobinuria), fever, back and abdominal pain, tachycardia; severe cases progress to acute kidney injury and death.
- Laboratory: fall in haemoglobin, raised unconjugated bilirubin and lactate dehydrogenase, reduced haptoglobin, Heinz bodies on supravital stain, 'bite cells' on the smear, reticulocytosis after recovery.
- Children aged 1 to 5 years are most affected. The same oxidant mechanism is triggered by primaquine, sulphonamides, naphthalene mothballs and infections, so favism is one member of a wider family of G6PD haemolytic crises.

3.4 Control

- Vicine and convicine are heat-stable, so ordinary cooking does not protect a G6PD-deficient person; soaking removes only part because the glycosides are only moderately water-soluble.
- Low-vicine faba bean genotypes (vc minus gene) carry less than 10 percent of normal levels.
- Dietary avoidance by affected individuals and neonatal G6PD screening are the practical controls.

[HPSC EXAM TRAP] Favism is not an allergy and not a cyanide poisoning. The bean is harmless to people with normal G6PD. A statement that 'faba beans are toxic to all consumers' is false; a statement that 'cooking abolishes favism' is also false.

4. Phytohaemagglutinins (Lectins)

4.1 Nature

- Lectins are carbohydrate-binding (glyco)proteins. The kidney bean lectin phytohaemagglutinin (PHA) of Phaseolus vulgaris (rajma, red kidney bean, also white kidney bean and cannellini) is a tetrameric protein of about 120 kilodaltons with two subunit types, PHA-E (erythroagglutinating) and PHA-L (leucoagglutinating).
- Raw red kidney beans contain 20,000 to 70,000 haemagglutinating units (hau) per gram; a properly boiled bean contains 200 to 400 hau. As few as 4 or 5 raw beans can cause illness.
- Related lectins: concanavalin A (jack bean), soybean agglutinin, ricin (castor bean; a ribosome-inactivating lectin, not a food lectin but the examiner's favourite comparator).

4.2 Mechanism

```
Ingested PHA survives gastric acid and proteolysis
        |
        v
Binds oligosaccharide receptors (N-acetylgalactosamine, galactose-rich glycans)
on the brush border of enterocytes
        |
        v
Disrupts microvilli, increases permeability, stimulates hyperplastic crypt
changes, impairs disaccharidase and transporter function
        |
        v
Malabsorption, bacterial overgrowth, fluid secretion, endocytosis of lectin
into the circulation (haemagglutination in vitro)
        |
        v
Nausea, vomiting within 1 to 3 hours, diarrhoea and abdominal pain within hours;
recovery usually in 3 to 4 hours, rarely needs hospitalisation
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- Chronic feeding of raw beans to animals causes growth depression, loss of nitrogen, pancreatic and intestinal hypertrophy.

4.3 Thermal destruction and the slow-cooker paradox

- Lectins are proteins and are denatured by moist heat at 100 degrees Celsius: soak dry beans at least 5 hours, discard the water, then boil vigorously for at least 10 minutes (30 minutes is the conservative recommendation). Pressure cooking is fully effective.
- Why slow cooking at sub-boiling temperatures (slow cookers at 70 to 80 degrees Celsius) increases toxicity: partial heating at these temperatures does not unfold PHA but increases its extraction and activity, so the cooked bean can show up to five times the haemagglutinating activity of the raw bean. Dry heat (roasting) and microwave heating of dry beans are also unreliable.
- Outbreaks in the United Kingdom were traced to raw or slow-cooked beans in salads and casseroles; canned kidney beans are already retorted and safe.

[HPSC EXAM TRAP] 'Any heating reduces lectin activity' is false. Only temperatures at or above boiling for an adequate time denature PHA; sub-boiling heating increases it. Also, lectin toxicity is a pure protein effect; it is not a cyanide effect and not an allergy.

5. Enzyme Inhibitors: Trypsin and Chymotrypsin Inhibitors

5.1 The two soybean inhibitor families

Property	Kunitz trypsin inhibitor (KTI)	Bowman-Birk inhibitor (BBI)
Molecular mass	About 20 to 21 kilodaltons (181 amino acids)	About 8 kilodaltons (71 amino acids)
Disulphide bonds	2	7 (very compact)
Reactive sites	Single-headed: one site, inhibits trypsin mainly (arginine 63 to isoleucine 64)	Double-headed: one site for trypsin (lysine 16), one for chymotrypsin (leucine 43)
Heat stability	Less stable; destroyed by moist heat	More stable; needs longer heating; survives dry heat
Acid and digestive stability	Partly degraded	Resistant to gastric acid and pepsin
Share of soy trypsin inhibitor activity	About 80 percent	About 20 percent

- Other legumes (chickpea, pigeon pea, kidney bean, lima bean, groundnut) contain mostly Bowman-Birk type inhibitors; potatoes and cereals contain other serine protease inhibitors.

5.2 Mechanism and pancreatic hypertrophy

Inhibitor forms a tight, slow-dissociating complex with trypsin in the duodenum
|
v
Free trypsin falls; protein digestion falls
|
v
Loss of trypsin-mediated negative feedback on the intestinal
CCK-releasing peptide (monitor peptide / luminal CCK-releasing factor)
|
v
More cholecystokinin (CCK) released from duodenal I cells
|
v
Pancreatic acinar cells chronically stimulated: hypersecretion of trypsinogen, chymotrypsinogen; acinar hypertrophy and hyperplasia (rodents); loss of sulphur amino acids in secreted enzymes; growth depression

- Because pancreatic enzymes are rich in cysteine and methionine, inhibitor-induced hypersecretion drains the limiting sulphur amino acids from a legume diet, aggravating protein deficiency.
- Rodents respond with pancreatic hypertrophy; humans and larger animals show hypersecretion but little hypertrophy, so the examiner expects the qualified statement.

5.3 Thermal inactivation dynamics

- Inactivation is first order in time and strongly moisture dependent: moist heat at 100 degrees Celsius for 15 to 20 minutes destroys 80 to 90 percent of soybean trypsin inhibitor activity; autoclaving at 120 degrees Celsius for 15 to 30 minutes removes virtually all. Dry roasting at the same temperature is far less effective.
- Over-heating causes Maillard loss of lysine, so toasting conditions are optimised to leave a small residual activity (about 10 to 20 percent of raw) while protecting protein quality.
- Analytical control: Trypsin Inhibitor Activity assay (benzoyl-DL-arginine-p-nitroanilide substrate), and the indirect urease activity test (change in pH of 0.05 to 0.30 units indicates adequately heated soy meal, because urease and trypsin inhibitor have similar heat sensitivity).

6. Mineral Chelators: Phytic Acid

6.1 The molecule

- Phytic acid is myo-inositol 1,2,3,4,5,6-hexakis(dihydrogen phosphate), abbreviated IP6, with six phosphate groups carrying up to 12 negative charges at intestinal pH.
- It is the principal phosphorus store of seeds: 1 to 5 percent of cereals, legumes, oilseeds and nuts by dry weight; in wheat and rice it is concentrated in the aleurone layer and bran, in legumes in the cotyledons, and in maize in the germ. Milling of wheat removes much of it; dehulling of legumes removes less.

6.2 Chelation mechanism

```
Phytate (12 negative charges) + divalent and trivalent cations
(Zn2+, Fe2+ / Fe3+, Ca2+, Mg2+, Cu2+)
      | at the near-neutral pH of the small intestine
      v
Insoluble metal-phytate and protein-metal-phytate ternary complexes
      |
      v
Minerals are not released at the mucosal surface; excreted in faeces
```

- Humans lack sufficient intestinal phytase, so the complexes pass unabsorbed. Zinc is the most affected mineral, iron next, then calcium and magnesium.
- Useful predictive ratios: a phytate-to-zinc molar ratio above 15 markedly impairs zinc absorption; a phytate-to-iron molar ratio above 1 impairs non-haem iron absorption; a calcium-times-phytate-to-zinc ratio above 200 is also inhibitory, because calcium potentiates zinc-phytate precipitation.
- Phytate also binds proteins (reducing digestibility) and starch, and inhibits alpha-amylase, trypsin and pepsin.

6.3 Degradation by phytase

- Phytase (myo-inositol hexakisphosphate phosphohydrolase) sequentially removes phosphate groups to give IP5, IP4, IP3 and ultimately inositol plus inorganic phosphate; only IP6 and IP5 are strongly inhibitory, so even partial dephosphorylation helps.
- Plant phytases are 6-phytases (cereals); microbial phytases are mostly 3-phytases (Aspergillus niger). Optimum pH 4.5 to 5.5, optimum temperature 45 to 55 degrees Celsius.

Process	Mechanism	Typical reduction of phytate
Soaking	Leaching and activation of endogenous phytase	20 to 50 percent
Germination (sprouting)	Strong synthesis and activation of phytase as the seed mobilises phosphorus	50 to 80 percent
Fermentation (idli, dosa, dhokla, sourdough, tempeh)	Microbial phytase plus acid pH optimum	50 to 90 percent
Cooking	Phytate is heat-stable; little reduction unless combined with soaking	Low
Milling	Physical removal of bran and germ	Large in wheat (80 percent in refined flour)

- Enhancers that counter phytate: ascorbic acid, citric acid and meat factors improve iron absorption even in the presence of phytate.
- Health note: phytate also has antioxidant and anti-carcinogenic properties in the colon, so a balanced answer mentions that complete elimination is not the goal in a mixed diet.

[HPSC EXAM TRAP] Phytic acid is an antinutrient, not a toxin; the harm is deficiency of zinc, iron and calcium, not direct injury. 'Cooking destroys phytate' is false; 'germination and fermentation reduce phytate' is true.

Part 1B: Other Botanical Toxicants

PART 1B: OTHER BOTANICAL TOXICANTS

1. Solanine and Chaconine: Glycoalkaloids of the Solanaceae

1.1 Chemistry and sources

- The two major potato (Solanum tuberosum) glycoalkaloids share the same steroidal aglycone, solanidine (a C27 cholestane-type alkaloid with a nitrogen-containing indolizidine ring). They differ in the trisaccharide at carbon 3:
  - alpha-Solanine: solatriose (glucose, galactose, rhamnose).
  - alpha-Chaconine: chacotriose (glucose and two rhamnose units).
- alpha-Chaconine is roughly 60 percent of the total and is the more toxic of the two; the pair act synergistically.
- Tomato contains alpha-tomatine (aglycone tomatidine), high in green fruit and degraded on ripening; brinjal contains solasonine and solamargine; nightshade berries contain solanine.
- Distribution in the potato: concentrated in the peel (30 to 80 percent of the total), in sprouts, eyes and green tissue; a normal tuber contains 20 to 100 mg/kg, a green, sprouted or damaged tuber may exceed 1,000 mg/kg.
- Greening is chlorophyll formation under light; the same light also stimulates glycoalkaloid synthesis, so greening is a marker, not the toxin itself.

1.2 Mechanism

```
Glycoalkaloid
   |
   +--> Inhibits acetylcholinesterase and butyrylcholinesterase (reversible)
   |        -> acetylcholine accumulates -> salivation, cramps, bradycardia,
   |           confusion, in severe cases respiratory depression
   |
   +--> Sugar side chain complexes with membrane 3-beta-hydroxysterols
            (cholesterol) -> membrane disruption, leakage
            -> gastrointestinal mucosal damage, haemolysis in vitro
```

- Absorption from the gut is poor, which is why gastrointestinal effects dominate and systemic cholinergic effects appear only at higher doses.

1.3 Clinical pathology

- Onset 8 to 12 hours (sometimes 30 minutes to 24 hours): nausea, vomiting, diarrhoea, abdominal cramps, burning throat, headache, fever; severe cases show drowsiness, hallucinations, bradycardia, hypotension, dilated pupils, respiratory failure.
- Toxic dose: 2 to 5 mg/kg body weight; potentially lethal at 3 to 6 mg/kg. Classic outbreak: 78 London schoolboys in 1979 from stored potatoes containing about 250 to 300 mg/kg.

1.4 Thermal stability and control

- Glycoalkaloids decompose only above 170 degrees Celsius (melting with decomposition near 260 degrees Celsius); boiling, baking and microwaving change them little, since they are not leached in quantity and not destroyed. Deep frying at 170 degrees Celsius or above reduces them partly because of thermal degradation and loss into oil.
- Practical control: store tubers in the dark, cool and dry; discard green, sprouted, bitter or damaged tubers; peel thickly (removes 30 to 80 percent); cut out eyes and sprouts.

- Safe ceiling: 200 mg total glycoalkaloids per kg fresh weight (20 mg/100 g), the value used by most regulators and plant breeders (new varieties exceeding it are not released); bitterness is detectable at about 140 mg/kg, burning at 200 mg/kg.

[HPSC EXAM TRAP] 'Boiling detoxifies green potatoes' is false. 'Peeling reduces glycoalkaloids' is true. 'The limit is 200 mg/kg for each of solanine and chaconine' is false: 200 mg/kg is the total.

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2. Goitrogens and Glucosinolates (Brassica family)

2.1 Chemistry

- Glucosinolates are beta-thioglucoside N-hydroxysulphates: a glucose attached through sulphur to a carbon that carries a variable side chain (R) and an N-O-sulphate group. More than 120 occur in the Brassicaceae (Cruciferae): cabbage, cauliflower, mustard, rapeseed, turnip, radish, kale, broccoli.

Glucosinolate	Side chain	Source	Hydrolysis product of concern
Progoitrin	2-Hydroxy-3-butenyl	Rapeseed, turnip, rutabaga, cabbage	Goitrin (5-vinyl-oxazolidine-2-thione)
Sinigrin	Allyl (2-propenyl)	Black mustard, cabbage, horseradish	Allyl isothiocyanate (pungency); thiocyanate ion
Glucobrassicin	Indol-3-ylmethyl	Cabbage, broccoli	Indole-3-carbinol and thiocyanate ion
Sinalbin	p-Hydroxybenzyl	White mustard	p-Hydroxybenzyl isothiocyanate

2.2 Myrosinase hydrolysis

```
Glucosinolate (in vacuole)  +  myrosinase (thioglucosidase, in separate 'myrosin' cells)
      | tissue disruption by chopping, chewing or insect attack
      v
Glucose + unstable thiohydroximate-O-sulphonate aglycone
      |
      +-- neutral pH -----> isothiocyanate (R-N=C=S)
      +-- acid pH, Fe2+ --> nitrile (R-C=N) and sulphur
      +-- indole or hydroxybenzyl side chain --> thiocyanate ion (SCN-)
      +-- 2-hydroxy side chain (progoitrin) --> isothiocyanate cyclises
                                                spontaneously to goitrin
                                                (oxazolidine-2-thione)
```

- Cooking inactivates myrosinase, but intact glucosinolates reaching the colon are hydrolysed by gut bacteria, so cooked cabbage is still mildly goitrogenic.

2.3 Two different antithyroid mechanisms (the examiner's distinction)

Agent	Site	Mechanism	Overcome by extra iodine?
Thiocyanate ion (SCN-) and other anions	Sodium-iodide symporter of the thyroid follicular cell	Competitive inhibition of iodide uptake, which the user's syllabus describes as competitive inhibition of thyroidal iodine uptake	Yes, high iodine intake restores uptake
Goitrin (oxazolidine-2-thione) and related thiones	Thyroid peroxidase (TPO)	Blocks organification (iodination of tyrosyl residues of thyroglobulin) and coupling, so thyroxine synthesis falls	No, iodine supplementation does not reverse it (thiourea-like action, similar to propylthiouracil)

- Consequences: lowered thyroxine and tri-iodothyronine, raised thyroid stimulating hormone, follicular hyperplasia and goitre. Human goitre from Brassica alone is rare; it becomes significant where iodine intake is marginal (the 'cabbage goitre' of the Himalayan and sub-Himalayan belts, and the cassava-thiocyanate goitre of Central Africa).
- Rapeseed meal for animals is limited to about 30 micromoles glucosinolate per gram; low-glucosinolate 'double-zero' canola (low erucic acid, low glucosinolate) was bred for this reason.
- Processing: blanching in boiling water leaches soluble glucosinolates (30 to 60 percent into the water) and inactivates myrosinase; fermentation (sauerkraut) hydrolyses them; adequate dietary iodine is the public-health control.

[HPSC EXAM TRAP] Goitrin is formed from progoitrin, not from sinigrin. Sinigrin gives allyl isothiocyanate (the hot taste of mustard). A pair that writes 'sinigrin to goitrin' is false.

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3. Oxalates

3.1 Forms and sources

- Oxalic acid (ethanedioic acid, HOOC-COOH) occurs as soluble sodium and potassium oxalate in cell sap, and as insoluble calcium oxalate crystals (raphides, needle-shaped bundles; druses; crystal sand).

Food	Approximate oxalate (mg/100 g fresh)	Predominant form
Spinach (palak)	600 to 1,300	Mainly soluble
Amaranth (chaulai)	500 to 1,000	Mainly soluble
Rhubarb (leaves far higher than stalks)	500 to 1,300 in stalks; leaves are poisonous	Both
Beet greens	600 to 900	Soluble
Colocasia (arbi, taro) leaves and corms	200 to 500, plus raphides	Insoluble raphides cause acridity
Tea (dry leaf), cocoa, nuts, star fruit	High	Soluble

3.2 Mechanism of harm

```
Soluble oxalate absorbed (2 to 15 percent of intake)
      |
      v
Binds ionised calcium in blood -> hypocalcaemia (acute, high dose only)
      -> tetany, convulsions, cardiac arrhythmia
      |
      v
Filtered by kidney, precipitates as calcium oxalate monohydrate
(whewellite) in tubules -> acute oxalate nephropathy; chronically,
nucleation of calcium oxalate renal calculi (about 75 percent of all
kidney stones)

Insoluble calcium oxalate (raphides) in colocasia and dieffenbachia
      |
      v
Needles pierce oral mucosa, with proteolytic enzyme co-released
      -> burning, itching, oedema of lips and tongue (acridity), rarely airway oedema
```

- In the gut, dietary oxalate binds calcium and lowers calcium absorption (antinutrient effect); conversely, dietary calcium taken with an oxalate-rich meal binds oxalate in the lumen and reduces oxalate absorption, which is why spinach with paneer or yoghurt is nutritionally rational.

- Acute lethal dose of oxalic acid is about 15 to 30 g in adults (not achievable from spinach, but achievable from rhubarb leaves or industrial oxalic acid, the classical 'salt of sorrel' poison).

3.3 Processing

- Boiling leaves in a large volume of water and discarding it removes 30 to 90 percent of soluble oxalate (depends on surface area and water volume); steaming removes less; frying removes almost none because there is no aqueous phase.
- Cooking colocasia with tamarind, lime or amchur (acid) and with calcium-rich ingredients reduces acridity; adequate boiling softens raphides.
- Stone patients are advised to limit high-oxalate foods, drink ample water and not to restrict dietary calcium.

[HPSC EXAM TRAP] Spinach is not a good calcium source despite its high calcium content, because the calcium is largely locked as oxalate. Rhubarb stalks are edible; rhubarb leaves are toxic. 'Frying removes oxalate' is false.

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4. Argemone mexicana and Epidemic Dropsy

4.1 The adulterant

- Argemone mexicana (prickly poppy, satyanashi, pili kateli, Mexican poppy) grows as a weed in mustard fields. Its seeds resemble black mustard (Brassica nigra) seeds in size and colour and are harvested with the crop or added deliberately because the seed oil is cheap.
- Argemone seeds contain 30 to 35 percent of a pale yellow oil rich in the benzophenanthridine (isoquinoline) alkaloids sanguinarine (about 0.3 percent of oil) and its reduced form dihydrosanguinarine (about 1 percent); the seed also contains protopine and berberine. Dihydrosanguinarine is converted to sanguinarine in the body.
- Dangerous level: oil containing about 1 percent argemone oil (0.01 percent sanguinarine) causes disease; outbreaks occur with 2 to 10 percent adulteration.

4.2 Mechanism

```
Sanguinarine (a quaternary benzophenanthridine cation)
      |
      +--> Inhibits Na+/K+-ATPase (ouabain-like) in vascular endothelium
      |      and cardiac muscle -> endothelial swelling, increased capillary
      |      permeability, plasma leakage
      |
      +--> Inhibits pyruvate oxidation and glucose-6-phosphatase, depletes
      |      hepatic glycogen; oxidative stress, lipid peroxidation
      |
      +--> Dilates capillaries and arterioles; pooling of blood
      |
      v
Non-inflammatory, bilateral pitting oedema of the legs, hypoproteinaemia,
high-output cardiac failure, retinal vascular dilatation and haemorrhages,
raised intra-ocular pressure (glaucoma)
```

4.3 Clinical pathology (epidemic dropsy)

- Onset 1 to 3 weeks after eating the adulterated oil; whole households fall ill together, which distinguishes it from cardiac or renal dropsy.
- Gastrointestinal prodrome: diarrhoea, vomiting, loss of appetite.
- Dropsy: bilateral pitting oedema of the lower limbs that spreads upwards, with erythema, tenderness and a sensation of heat; dilated superficial veins; sarcoid-like skin nodules; hair loss.

- Cardiovascular: tachycardia, cardiomegaly, congestive high-output cardiac failure (the main cause of death, mortality 5 percent in outbreaks).
- Eyes: glaucoma from raised intra-ocular pressure, retinal haemorrhage and venous dilatation, blurred vision; can cause blindness.
- Others: mild anaemia, fever, breathlessness, proteinuria, hepatomegaly.
- Major outbreaks: Calcutta 1877 (first description); Delhi 1998 (about 3,000 cases, more than 60 deaths, which triggered the ban on loose sale of edible oil); recurring episodes in Bihar, Uttar Pradesh, Madhya Pradesh, Gwalior 2000 and 2012.
- Treatment is supportive (diuretics, calcium, antioxidants, management of glaucoma); removal of the oil from the diet is curative in survivors.

4.4 Analytical identification

Test	Principle	Procedure	Sensitivity
Nitric acid test (Hofmann test, the official qualitative test)	Sanguinarine forms a coloured (red to orange-red) complex with concentrated nitric acid	Shake 5 mL oil with 5 mL concentrated nitric acid in a test tube; a reddish-brown to orange-red colour in the acid layer indicates argemone oil	Detects about 0.25 percent and above
Ferric chloride test	Sanguinarine gives an orange-brown precipitate	Shake oil with 10 mL ferric chloride solution, heat in a boiling water bath; precipitate or dark colour indicates argemone	About 1 percent
Paper chromatography (confirmatory official method)	Separation of sanguinarine and dihydrosanguinarine from the oil matrix; the spots fluoresce orange-yellow under ultraviolet light; comparison of Rf with authentic sanguinarine	Extract oil with ethanol, concentrate, spot on Whatman paper, develop, view under ultraviolet	Detects as low as 0.0001 percent (1 ppm)
Thin-layer chromatography and high-performance liquid chromatography	Quantitative estimation of sanguinarine and dihydrosanguinarine	Silica gel plate or reversed-phase column, fluorescence or ultraviolet detection	Quantitative confirmation, microgram levels
Physical examination of seed	Argemone seed is black, rough, with a reticulated (net-like) surface, slightly larger than mustard, bitter, and shows a white kernel inside when crushed; mustard seed is smooth with a yellowish kernel	Hand lens and crushing test	Field screening

4.5 Statutory position

- Food Safety and Standards (Food Products Standards and Food Additives) Regulations, 2011, standard for mustard oil and all edible oils: the test for argemone oil shall be negative; argemone oil is a prohibited adulterant with nil tolerance.
- Food Safety and Standards (Prohibition and Restrictions on Sales) Regulations, 2011: prohibition on sale of edible oil adulterated with argemone oil and, since 1998, restriction on sale of loose edible oil (packaged oil must carry the manufacturer's identification).
- Under the Act, argemone adulteration that causes injury is treated as unsafe food attracting the severest penalties (Section 59, up to life imprisonment where death results).

[HPSC EXAM TRAP] Epidemic dropsy is not a nutritional oedema (not protein-energy malnutrition) and not a cardiac disease of the individual; it is a poisoning that affects a whole family. The screening test is nitric acid (0.25 percent); the confirmatory and most sensitive test is paper chromatography (0.0001 percent). A pair reversing these sensitivities is false.

Part 2: Marine & Animal Biotoxins

PART 2: MARINE AND ANIMAL BIOTOXINS

First-principles orientation

Most marine biotoxins are made not by the animal eaten but by microscopic planktonic algae (dinoflagellates and diatoms) or by bacteria. Filter-feeding bivalves (mussels, clams, oysters, scallops) concentrate the algae; reef fish concentrate algal toxins through the food chain; pufferfish concentrate bacterial toxin. Nearly all of these toxins are small, water-soluble or lipid-soluble molecules that are heat-stable, tasteless and odourless, so cooking is useless and the only control is monitoring of harvesting waters and the product. The majority act on ion channels or signalling enzymes of nerve and muscle, which is why the clinical pictures are neurological.

Shared background: the voltage-gated sodium channel

- Nerve conduction and muscle contraction depend on the rapid opening of voltage-gated sodium channels (Nav). Toxins bind at numbered receptor sites:
  - Site 1 (outer pore mouth): saxitoxin and tetrodotoxin plug the pore and block sodium entry; the nerve cannot fire; result is flaccid paralysis.
  - Site 5 (transmembrane segments): brevetoxins and ciguatoxins bind, shift activation to more negative potentials and prevent inactivation; the channel opens persistently; result is repetitive firing, paraesthesiae and the temperature-reversal sensation.

---

1. Shellfish Poisoning Syndromes

1.1 Paralytic Shellfish Poisoning (PSP)

Feature	Detail
Toxins	Saxitoxin (STX) and more than 50 analogues (gonyautoxins, neosaxitoxin, decarbamoyl and sulphocarbamoyl derivatives); saxitoxin is a tricyclic tetrahydropurine with two guanidinium groups, water-soluble, heat-stable, acid-stable, destroyed by alkali
Producers	Marine dinoflagellates Alexandrium (catenella, tamarense, minutum; formerly Gonyaulax), Gymnodinium catenatum, Pyrodinium bahamense; also freshwater cyanobacteria (Anabaena, Aphanizomenon)
Vectors	Mussels, clams, oysters, scallops, cockles; also crabs, lobsters, gastropods that feed on contaminated bivalves
Mechanism	Reversible, high-affinity block of site 1 of the voltage-gated sodium channel of nerve and skeletal muscle; conduction fails; neuromuscular transmission and respiratory muscle contraction stop; cardiac muscle is relatively spared
Potency	One of the most potent known non-protein toxins: lethal oral dose about 1 to 4 mg (about 10 to 20 micrograms per kg); 5 to 30 micrograms per kg causes symptoms
Clinical picture	Onset 5 to 30 minutes, up to 2 hours: tingling and numbness of lips, tongue and face spreading to limbs; floating sensation, headache, dizziness, incoordination, dysarthria; in severe cases muscle weakness progressing to respiratory paralysis within 2 to 12 hours; consciousness preserved; mortality 1 to 15 percent, mainly within 12 hours if ventilation is not provided; survivors recover fully
Treatment	No antidote; mechanical ventilation, gastric lavage, activated

Feature	Detail
	charcoal, supportive care
Regulatory action level	80 micrograms STX equivalents per 100 g shellfish meat (0.8 mg/kg), the Codex and worldwide standard
Analysis	Mouse bioassay (AOAC official method; 1 mouse unit = dose killing a 20 g mouse in 15 minutes), replaced progressively by high-performance liquid chromatography with pre- or post-column oxidation and fluorescence detection (Lawrence method), liquid chromatography-mass spectrometry, receptor-binding assay

1.2 Diarrhetic Shellfish Poisoning (DSP)

Feature	Detail
Toxins	Okadaic acid (OA) and dinophysistoxins (DTX-1, DTX-2, DTX-3), lipophilic polyether fatty acids; heat-stable; often co-occur with pectenotoxins and yessotoxins
Producers	Dinoflagellates Dinophysis (acuta, fortii, acuminata) and benthic Prorocentrum lima
Vectors	Mussels, scallops, clams; toxin concentrates in the hepatopancreas (digestive gland)
Mechanism	Potent, specific inhibition of the serine/threonine protein phosphatases PP2A (most strongly) and PP1; proteins that control cytoskeleton, ion transport and secretion remain hyperphosphorylated; in enterocytes this causes loss of tight-junction integrity, increased paracellular permeability and secretion of fluid and electrolytes; okadaic acid is also a tumour promoter (phosphatase inhibition activates proliferation pathways)
Clinical picture	Onset 30 minutes to a few hours (usually within 12 hours): profuse diarrhoea, nausea, vomiting, abdominal pain, chills; self-limiting within 3 days; no fatalities recorded; chronic exposure suspected to promote gastrointestinal tumours
Regulatory limit	European Union: 160 micrograms okadaic acid equivalents per kg whole flesh; Codex: 0.16 mg/kg
Analysis	Mouse bioassay (historical), protein phosphatase inhibition assay, enzyme-linked immunosorbent assay, liquid chromatography-mass spectrometry (reference method)

1.3 Neurotoxic Shellfish Poisoning (NSP)

Feature	Detail
Toxins	Brevetoxins (PbTx-1 to PbTx-10), lipid-soluble cyclic polyether ladder molecules; heat-stable
Producers	Dinoflagellate Karenia brevis (formerly Gymnodinium breve, Ptychodiscus brevis), the Florida and Gulf of Mexico 'red tide' organism; also New Zealand
Vectors	Oysters, clams, mussels; unusual feature: aerosolised toxin in sea spray during blooms causes respiratory irritation on beaches; also massive fish and manatee kills
Mechanism	Binds site 5 of the voltage-gated sodium channel; opens the channel at resting potential and blocks inactivation; persistent sodium influx, membrane depolarisation and repetitive firing; release of acetylcholine at the neuromuscular junction
Clinical picture	Onset 30 minutes to 3 hours: paraesthesiae of mouth and limbs, temperature reversal sensation (shared with ciguatera), gastrointestinal upset, ataxia, dizziness, bradycardia or tachycardia, dilated pupils; no deaths recorded; recovery within days; inhalation

Feature	Detail
	causes cough, bronchoconstriction, conjunctival irritation
Regulatory limit	20 mouse units per 100 g (about 0.8 mg PbTx-2 equivalents per kg) in the United States
Analysis	Mouse bioassay, enzyme-linked immunosorbent assay, receptor-binding assay, liquid chromatography-mass spectrometry

1.4 Amnesic Shellfish Poisoning (ASP)

Feature	Detail
Toxin	Domoic acid, a water-soluble tricarboxylic amino acid of the kainoid class (structural analogue of glutamate and kainic acid); heat-stable
Producer	Diatoms, not dinoflagellates: Pseudo-nitzschia (multiseries, australis, pungens); also the red alga Chondria armata (from which it was first isolated as an anthelmintic in Japan)
Vectors	Mussels, razor clams, scallops, crabs; also anchovies and sardines, through which sea lions and sea birds are poisoned
Mechanism	Potent agonist at ionotropic glutamate receptors, particularly kainate and AMPA receptors; sustained excitation, calcium overload and excitotoxic neuronal death, concentrated in the hippocampus (CA1 and CA3 regions) and amygdala, the brain regions responsible for forming new memories; the blood-brain barrier is crossed poorly, so the very young and the elderly with leakier barriers are most affected
Clinical picture	Onset 15 minutes to 38 hours: vomiting, diarrhoea, abdominal cramps; then headache, confusion, disorientation, seizures, coma; survivors show permanent anterograde amnesia (loss of short-term memory), peripheral neuropathy and seizures; first recognised in the Prince Edward Island (Canada) outbreak of 1987 from blue mussels: 107 cases, 3 deaths, 12 with lasting memory loss
Regulatory limit	20 mg domoic acid per kg (20 micrograms per gram) of shellfish meat, Codex and worldwide
Analysis	High-performance liquid chromatography with ultraviolet detection at 242 nm (the AOAC reference method), liquid chromatography-mass spectrometry, enzyme-linked immunosorbent assay; the mouse bioassay is insensitive

[HPSC EXAM TRAP] Three traps in one table: (1) ASP is from a diatom, the other three are from dinoflagellates; (2) saxitoxin blocks the sodium channel, brevetoxin opens it; (3) okadaic acid does not act on an ion channel at all, it inhibits protein phosphatases. A fourth: PSP and ASP limits are both expressed per mass but in different units, 80 micrograms per 100 g versus 20 mg per kg.

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2. Finfish Toxins

2.1 Ciguatera Fish Poisoning (CFP)

Feature	Detail
Toxins	Ciguatoxins (CTX; Pacific P-CTX-1 is the most potent, Caribbean C-CTX-1), lipid-soluble cyclic polyether ladders; maitotoxin (MTX), the largest and most potent non-protein toxin known, water-soluble; also scaritoxin and gambierol
Producer	Benthic dinoflagellate Gambierdiscus toxicus and related species living as epiphytes on dead coral and macroalgae in tropical reefs

Feature	Detail
	between 35 degrees north and 35 degrees south latitude
Bioaccumulation chain	Gambierdiscus produces gambiertoxins; herbivorous reef fish (surgeonfish, parrotfish) graze the algae; carnivorous reef fish eat the herbivores; toxins are oxidised in the fish liver into more potent ciguatoxins and biomagnified; the largest, oldest predators are most toxic: barracuda, grouper, red snapper, amberjack, moray eel, Spanish mackerel, coral trout; viscera (liver, roe) carry most toxin
Mechanism	Ciguatoxin binds site 5 of the voltage-gated sodium channel (same site as brevetoxin but with higher affinity), causing persistent activation at resting potential, spontaneous and repetitive action potentials in sensory nerves and swelling of nodes of Ranvier; maitotoxin activates non-selective cation channels and calcium entry, causing calcium-dependent cell damage
Clinical picture	Onset 1 to 24 hours (usually 2 to 6 hours). Gastrointestinal phase: vomiting, diarrhoea, cramps. Neurological phase: circumoral and extremity paraesthesiae, pruritus, myalgia, arthralgia, and the pathognomonic temperature reversal dysaesthesia or cold allodynia (cold objects feel hot or burning and hot feels cold, because persistently activated cold-sensitive fibres misfire); metallic taste, dental pain, blurred vision. Cardiovascular: bradycardia, hypotension. Symptoms persist for weeks to months and may be reactivated by alcohol, nuts or fish; mortality below 0.5 percent; ciguatera is the commonest non-bacterial fish-borne illness worldwide (10,000 to 50,000 cases per year)
Control	Toxins are heat-stable, freeze-stable, tasteless and not removed by drying, salting or smoking; no visual or sensory test; avoid large predatory reef fish and their viscera; mouse bioassay, receptor-binding assay, cytotoxicity assay and liquid chromatography-mass spectrometry; United States guidance level 0.01 micrograms P-CTX-1 equivalents per kg fish; intravenous mannitol has been used empirically for neurological symptoms

2.2 Scombroid (Histamine) Poisoning

Feature	Detail
Fish involved	Dark-fleshed, fast-swimming fish with high free histidine in muscle: family Scombridae (tuna, mackerel, bonito, skipjack, seer fish) and non-scombroid species such as sardine, anchovy, mahi-mahi (dolphin fish), bluefish, herring, marlin
Chemistry	Free L-histidine is decarboxylated to histamine by bacterial histidine decarboxylase (a pyridoxal-phosphate dependent enzyme); the bacteria are normal gut and gill flora, mainly Gram-negative enteric organisms: Morganella morganii (the strongest producer), Klebsiella pneumoniae, Hafnia alvei, Enterobacter aerogenes, Proteus, Raoultella; also Photobacterium at low temperature; fastest at 20 to 30 degrees Celsius; production is negligible below 4 degrees Celsius but, once formed, histamine and the enzyme persist
Why it is a 'pseudo-allergy'	Histamine acts directly on H1 and H2 receptors of blood vessels, bronchi, gut and skin, reproducing the picture of an immunoglobulin E mediated allergic reaction without any antibody; it occurs in people with no fish allergy, affects several people eating the same fish, and responds to antihistamines; co-amines in spoiled fish (cadaverine, putrescine) inhibit intestinal diamine oxidase and potentiate histamine; this is why gut-wall inactivation fails and why patients on isoniazid or monoamine oxidase inhibitors are more sensitive
Clinical picture	Onset 10 minutes to 2 hours: flushing of face and neck, urticaria, pruritus, headache, palpitations, burning or peppery taste, oral tingling, nausea, diarrhoea, abdominal cramps, hypotension, bronchospasm; resolves within 12 to 24 hours; rarely fatal; treated with antihistamines (H1 and H2 blockers)

Feature	Detail
Thermal stability	Histamine is heat-stable: cooking, canning, smoking, freezing and retorting do not destroy it; canned tuna is a classic vehicle; sensory spoilage signs may be absent, so the fish can taste normal (sometimes peppery or honeycombed flesh)
Regulatory limits	Codex Standard for fish and fishery products: no sample above 100 mg/kg histamine as a hygiene indicator; 200 mg/kg as the decomposition limit; European Union: two-class sampling plan with 100 mg/kg (m) and 200 mg/kg (M) in fresh fish, 200 and 400 mg/kg in enzyme-ripened products; United States Food and Drug Administration action level 50 mg/kg (because this level indicates that parts of the lot may exceed 500 mg/kg); FSSAI standard for fish and fishery products specifies a histamine ceiling of 200 mg/kg for histamine-prone species (confirm against the current gazette)
Analysis	AOAC fluorometric method (histamine condensed with o-phthaldialdehyde), enzyme-linked immunosorbent assay rapid kits, high-performance liquid chromatography; prevention by rapid chilling below 4 degrees Celsius within hours of catch (time-temperature control, the critical control point in the HACCP plan for tuna)

2.3 Tetrodotoxin (TTX)

Feature	Detail
Source organisms	Pufferfish (fugu) of the family Tetraodontidae (Takifugu, Fugu, Lagocephalus), porcupine fish (Diodontidae), ocean sunfish, blue-ringed octopus, certain newts (Taricha), toads (Atelopus), xanthid crabs, horseshoe crab eggs, some gastropods
Origin of the toxin	Produced by symbiotic and gut bacteria (Vibrio alginolyticus, Vibrio fischeri, Pseudomonas, Shewanella, Alteromonas, Bacillus) and accumulated through the food chain; cultured pufferfish fed a toxin-free diet are non-toxic, which proves the exogenous origin; the fish is resistant because its own sodium channels carry a mutation (aromatic residue substitution in domain I) that prevents toxin binding
Distribution in the fish	Highest in ovary and liver, then intestine, skin; muscle of most species is nearly toxin-free, which is the basis of licensed fugu preparation in Japan
Chemistry	Aminoperhydroquinazoline with a guanidinium group and a hemilactal; heat-stable, water-soluble, stable to cooking but destroyed by strong alkali
Mechanism	Binds site 1 of the voltage-gated sodium channel from the outside and plugs the pore (the guanidinium group mimics the hydrated sodium ion); pharmacologically identical to saxitoxin; blocks conduction in sensory and motor nerves and skeletal muscle; does not cross the blood-brain barrier well, so consciousness is retained
Clinical picture	Onset 10 to 45 minutes (up to 3 hours): perioral numbness and tingling; salivation, nausea; then ascending flaccid paralysis: limbs, trunk, respiratory muscles; aphonia, dysphagia, fixed dilated pupils, hypotension, bradycardia; death by respiratory failure within 4 to 6 hours in 50 to 60 percent of untreated severe cases; patients may appear dead while fully conscious ('zombie' phenomenon); lethal dose about 1 to 2 mg, about 10 times more toxic than cyanide by weight; four grades of severity are used clinically
Treatment	No antidote; early gastric lavage, activated charcoal, mechanical ventilation; if ventilated through the first 24 hours, recovery is complete
Control	In India, sale of pufferfish for food is not standardised and is effectively prohibited; Japan licenses fugu chefs who remove ovary, liver and skin; mouse bioassay (Japanese official method; limit 10 mouse units per gram, about 2 mg/kg), high-performance liquid

Feature	Detail
	chromatography with post-column fluorescence, liquid chromatography-mass spectrometry

[HPSC EXAM TRAP] Tetrodotoxin and saxitoxin have the same mechanism (site 1 block) but different origins: tetrodotoxin is bacterial via the pufferfish, saxitoxin is dinoflagellate via shellfish. 'Tetrodotoxin is produced by the pufferfish itself' is false. 'Scombroid poisoning can be prevented by cooking' is false; it is prevented by chilling the fish quickly after catch. 'Ciguatera fish look or taste abnormal' is false.

Part 3A: Heavy Metals

PART 3A: HEAVY METALS AND TOXIC TRACE ELEMENTS

First-principles orientation

Heavy metals are not metabolised and not destroyed by cooking; the body can only excrete them slowly. Harm therefore depends on cumulative body burden, on the chemical species (ionic, organic, valence) that determines absorption and target organ, and on the age of the consumer (children absorb more lead, foetuses are more sensitive to methylmercury). Three mechanisms recur: (1) binding to sulphydryl groups of enzymes, (2) displacement of essential metals (lead for calcium and zinc, cadmium for zinc and calcium, arsenate for phosphate), and (3) oxidative stress. The analytical backbone is the same for all: acid digestion (microwave) followed by atomic absorption spectrometry (flame, graphite furnace, hydride generation for arsenic, cold vapour for mercury) or inductively coupled plasma mass spectrometry.

Statutory framework

- Food Safety and Standards (Contaminants, Toxins and Residues) Regulations, 2011, Regulation 2.1 (Metal Contaminants), as harmonised with Codex by the 2018 to 2020 amendments: commodity-specific maximum limits in mg/kg (parts per million) for lead, cadmium, mercury, methylmercury, arsenic, tin, copper, zinc, nickel and chromium.
- Section 3(1)(zz) of the Act: food is unsafe if it contains contaminants beyond specified limits.
- Codex General Standard for Contaminants and Toxins in Food and Feed (CXS 193-1995) is the international reference; JECFA sets Provisional Tolerable Weekly Intakes (PTWI) or Provisional Tolerable Monthly Intakes (PTMI).

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1. Lead (Pb)

1.1 Sources in food

- Leaded petrol (historical), lead-based paints, lead-soldered cans, lead-glazed ceramics (acidic foods such as pickles, fruit juices and curd leach lead), brass and bronze utensils, lead water pipes, contaminated soil and irrigation water near smelters and battery recycling, lead chromate adulteration of turmeric and spices, calcium supplements from bone meal, and dust deposition on leafy vegetables grown by roadsides.
- Children absorb 40 to 50 percent of ingested lead; adults 10 to 15 percent; absorption rises with iron, calcium and zinc deficiency and with fasting.

1.2 Mechanism: the haem synthesis block

```
Succinyl-CoA + glycine
      | delta-aminolevulinic acid synthase (mitochondria)
      v
Delta-aminolevulinic acid (ALA)  <-- accumulates and spills into urine (neurotoxic itself)
      | delta-aminolevulinic acid dehydratase (ALAD), a zinc enzyme
      | *** INHIBITED BY LEAD (lead displaces zinc from the active-site cysteines) ***
      v
Porphobilinogen -> ... -> coproporphyrinogen III (coproporphyrin rises in urine)
      |
      v
Protoporphyrin IX
      | ferrochelatase (mitochondrial, inserts Fe2+)
      | *** INHIBITED BY LEAD ***
```

```
      v
Haem  <-- falls; zinc protoporphyrin (ZPP) and free erythrocyte protoporphyrin rise
```

- Result: microcytic hypochromic anaemia (haem synthesis fails despite adequate iron), shortened red cell survival, and basophilic stippling of red cells (lead inhibits pyrimidine 5'-nucleotidase, so ribosomal RNA fragments persist).
- ALAD inhibition is the most sensitive biochemical marker; zinc protoporphyrin is the screening marker; blood lead is the definitive measure.

1.3 Neurological and other toxicity

- Lead mimics calcium: it enters neurons through calcium channels, interferes with neurotransmitter release, activates protein kinase C, and disrupts the blood-brain barrier.
- Children: irreversible loss of intelligence quotient (about 1 to 3 points per 10 micrograms per decilitre at low levels), behavioural problems, hearing loss, growth retardation; acute encephalopathy with cerebral oedema and convulsions above about 70 micrograms per decilitre. No threshold is considered safe; the United States reference value for action is 3.5 micrograms per decilitre.
- Adults: peripheral motor neuropathy (wrist drop, foot drop), lead colic (abdominal pain, constipation), Burton's blue line on the gums, nephropathy (proximal tubular damage, gout, chronic kidney disease), hypertension, reduced fertility, and in pregnancy, miscarriage and low birth weight; lead stored in bone is mobilised during pregnancy and lactation.
- JECFA withdrew the PTWI of 25 micrograms per kg body weight in 2010 because no safe level could be identified.
- Treatment: remove the source; chelation with calcium disodium edetate, dimercaprol or succimer (dimercaptosuccinic acid) in symptomatic or high-level cases.

1.4 Representative FSSAI limits for lead (mg/kg)

Food category	Limit
Milk; infant formula and infant foods	0.02
Cereal grains; pulses	0.2
Fruits; fruiting and root vegetables	0.1
Leafy vegetables (brassicas)	0.3
Edible fats and oils	0.1
Fish	0.3
Crustaceans	0.5
Bivalve molluscs; cephalopods	1.0 (cephalopods) and 1.5 (bivalves)
Fruit juices and nectars	0.05
Table salt (iodised and iron-fortified)	2.0
Tea; turmeric (whole and powder); spices (dried)	10
Packaged drinking water and natural mineral water	0.01 mg/L
Foods not otherwise specified	2.5 (older general ceiling; verify current status)

2. Cadmium (Cd)

2.1 Sources

- Phosphate fertilisers (cadmium is a natural contaminant of rock phosphate), sewage sludge, zinc and lead smelting and mining effluent, electroplating, nickel-cadmium batteries, cigarette smoke (1 to 2 micrograms per cigarette; smokers have double the body burden).
- Plants take up cadmium readily: rice (especially in flooded paddies irrigated with contaminated water), wheat, leafy vegetables, potatoes; the richest animal sources are kidney and liver, and the hepatopancreas of crustaceans and cephalopods (crab brown meat, squid), and oysters and mussels. Cocoa and chocolate are also notable.

2.2 Mechanism

Ingested Cd2+ (absorption 3 to 5 percent, higher with iron deficiency, as cadmium uses the divalent metal transporter DMT1)
|
v
Liver synthesises metallothionein (cysteine-rich, binds Cd as Cd-MT)
|
v
Cd-MT released into blood, freely filtered by the glomerulus, reabsorbed by proximal tubular cells, degraded in lysosomes, releasing free Cd2+
|
v
When tubular metallothionein capacity is exceeded (renal cortex above about 200 micrograms per gram), free Cd2+ binds sulphydryl enzymes, generates oxidative stress and triggers apoptosis of proximal tubular cells
|
v
Irreversible proximal tubular dysfunction: low-molecular-weight proteinuria (beta-2-microglobulin, retinol-binding protein), glucosuria, aminoaciduria, phosphaturia, calciuria (a Fanconi-like syndrome)
|
v
Loss of phosphate and calcium, reduced renal activation of vitamin D (1-alpha-hydroxylase), plus direct cadmium inhibition of osteoblasts and substitution for calcium in bone mineral
|
v
Osteomalacia and osteoporosis, bone demineralisation, multiple pseudofractures

- Biological half-life is 10 to 30 years; the kidney and liver hold about half the body burden; the critical organ is the renal cortex.
- Cadmium is classified as a Group 1 human carcinogen (lung, prostate, kidney) by the International Agency for Research on Cancer.
- JECFA PTMI: 25 micrograms per kg body weight per month (set in 2010, replacing the PTWI of 7 micrograms per kg per week).

2.3 Itai-Itai disease

- 'Itai-itai' means 'ouch-ouch', from the cries of pain of patients. Occurred in the Jinzu River basin, Toyama Prefecture, Japan, from the 1910s and recognised in 1955 to 1968; caused by cadmium discharged from the Mitsui Mining and Smelting zinc mine at Kamioka into irrigation water, concentrated in rice.
- Victims were mainly post-menopausal multiparous women with low calcium and vitamin D status: severe bone pain, waddling gait, spontaneous fractures of ribs and pelvis, skeletal deformity, height loss, renal tubular dysfunction, anaemia; many were bedridden.

- It was the first case of pollution-related disease legally recognised in Japan (1968), the lesson being that environmental cadmium reaches humans through the staple crop.

2.4 Representative FSSAI limits for cadmium (mg/kg)

Food category	Limit
Rice (polished)	0.1 (Indian value; Codex allows 0.4; verify current notification)
Wheat; cereal grains other than rice and wheat	0.1
Pulses	0.1
Leafy vegetables; celery; fresh herbs	0.2
Root and tuber vegetables (potato peeled)	0.1
Fruits	0.05
Fish	0.3
Crustaceans	0.5
Cephalopods (without viscera)	2.0
Bivalve molluscs	2.0
Edible salt	0.5
Packaged drinking water	0.003 mg/L

---

3. Mercury (Hg)

3.1 Three chemical forms, three toxicities

Form	Route and absorption	Target	Example
Elemental (metallic) mercury, Hg0	Vapour inhaled, 80 percent absorbed by lung; almost nil absorbed from the gut if swallowed (less than 0.01 percent), so a swallowed thermometer bulb is harmless	Brain (crosses blood-brain barrier as vapour, oxidised to Hg2+ by catalase and trapped), kidney	Dental amalgam, thermometers, chlor-alkali plants, gold panning
Inorganic mercury salts, Hg2+ (mercuric chloride)	7 to 15 percent absorbed from gut; corrosive	Kidney (proximal tubule), gut	Skin-lightening creams, disinfectants, 'calomel' teething powders (historical pink disease, acrodynia)
Organic mercury, methylmercury CH3Hg+	95 percent absorbed from gut; lipophilic; bound to cysteine it mimics methionine and crosses the blood-brain barrier and the placenta by the large neutral amino acid transporter	Central nervous system, especially cerebellum, visual cortex, developing foetal brain	Fish and shellfish, fungicide-treated seed grain (Iraq 1971 to 1972, 6,500 cases)

3.2 How fish become the source: biomethylation and biomagnification

```
Inorganic mercury from industrial effluent, coal combustion, gold mining,
volcanoes deposited in water and sediment
      | sulphate-reducing and iron-reducing bacteria in anoxic sediment
      v
Methylmercury (CH3Hg+), lipophilic, binds tightly to cysteine in fish muscle protein
      |
      v
Plankton -> small fish -> larger fish -> top predators
```

(biomagnification by a factor of 10 to 100,000; mercury in fish is 90 to 100 percent methylmercury and cannot be removed by cooking, trimming or washing)
|
v
Highest in large, long-lived, predatory, pelagic fish: shark, swordfish, king mackerel, tilefish, bigeye tuna, marlin; lowest in sardine, anchovy, pomfret, small mackerel, shrimp, salmon, tilapia

3.3 Mechanism

- Methylmercury has extreme affinity for sulphydryl groups; it binds tubulin and disrupts microtubules (so neuronal migration in the foetal brain fails), inhibits protein synthesis, depletes glutathione, causes oxidative damage and disturbs calcium homeostasis and neurotransmitter release.
- The adult brain lesion is focal (cerebellar granule cells, calcarine visual cortex, dorsal root ganglia); the foetal lesion is diffuse because migration is disrupted everywhere.

3.4 Clinical pathology and Minamata disease

- Adult methylmercury poisoning (Hunter-Russell syndrome, 1940): paraesthesiae of lips and fingers, constriction of visual fields (tunnel vision), cerebellar ataxia, dysarthria, deafness, tremor, cognitive decline; latency weeks to months, dose-dependent.
- Minamata disease: Minamata Bay, Kumamoto, Japan; the Chisso Corporation's acetaldehyde plant used mercuric sulphate as catalyst and discharged methylmercury in effluent from 1932 to 1968; fish and shellfish in the bay accumulated it; the disease was recognised in 1956 (cats 'dancing' and dying first); more than 2,200 officially certified patients, more than 1,700 deaths, and thousands more affected; a second outbreak at Niigata (Agano River, Showa Denko) in 1965.
- Congenital (foetal) Minamata disease: mothers with little or no symptoms gave birth to children with cerebral palsy-like syndromes, microcephaly, mental retardation, seizures, blindness and deafness, proving that the foetus is far more sensitive than the mother; this is the basis of fish-consumption advisories for pregnant women.
- JECFA PTWI for methylmercury: 1.6 micrograms per kg body weight per week (foetal protection); for total mercury 4 micrograms per kg per week.
- Minamata Convention on Mercury (2013, in force 2017; India ratified in 2018) phases out mercury products and controls emissions.

3.5 Representative FSSAI limits for mercury (mg/kg)

Food category	Limit
Fish (general)	0.5 total mercury
Large predatory fish (shark, swordfish, tuna, marlin, pike)	1.0 methylmercury
Other foods (general, older ceiling)	1.0 (verify current specification)
Packaged drinking water	0.001 mg/L
Table salt and food additives	0.1 (as specified in individual standards)

---

4. Arsenic (As)

4.1 Species and sources

Species	Toxicity	Occurrence
Inorganic arsenite, As(III), trivalent	Most toxic (about 60 times more than As(V)); binds vicinal dithiols	Groundwater, contaminated rice, seaweed hijiki
Inorganic arsenate, As(V), pentavalent	Toxic; phosphate analogue	Groundwater, soil, pesticide-treated land (lead arsenate, historical)

Species	Toxicity	Occurrence
Methylated metabolites (monomethylarsonic acid, dimethylarsinic acid)	Intermediate; trivalent methylated forms are highly toxic	Formed in liver; also in rice
Arsenobetaine and arsenosugars	Essentially non-toxic, excreted unchanged	Fish and shellfish (this is why total arsenic in fish is high but harmless, and why limits for seafood refer to inorganic arsenic or are set higher)

- Groundwater: the Ganga-Brahmaputra-Meghna delta is the world's largest arsenic calamity: West Bengal (Murshidabad, Nadia, North and South 24 Parganas), Bihar, Uttar Pradesh, Assam, Jharkhand and Bangladesh, where shallow tube wells tap arsenic-rich alluvial aquifers (geogenic release from iron oxyhydroxides under reducing conditions).
- Rice: grown in flooded (anaerobic) paddies, which keep arsenic in the mobile As(III) form; rice roots take up arsenite through the silicon transporters Lsi1 and Lsi2, so rice accumulates ten times more arsenic than other cereals; bran and husk contain the most, so brown rice has more than polished rice; cooking in excess arsenic-free water and discarding it removes 30 to 60 percent, but cooking in arsenic-contaminated water adds arsenic.

4.2 Mechanism

```
Arsenite, As(III)
      | binds vicinal (paired) sulphydryl groups of dihydrolipoamide in the
      | pyruvate dehydrogenase and alpha-ketoglutarate dehydrogenase complexes
      v
Blocks entry of pyruvate into the citric acid cycle; ATP production falls;
also binds keratin (hair, nails, skin), inhibits glutathione reductase and
thioredoxin reductase, generates reactive oxygen species, inhibits DNA repair

Arsenate, As(V)
      | chemically mimics phosphate (PO4 3-)
      v
Substitutes for phosphate in glycolysis (1-arseno-3-phosphoglycerate forms
instead of 1,3-bisphosphoglycerate and hydrolyses spontaneously: 'arsenolysis'),
uncoupling ATP synthesis; is reduced in vivo to arsenite
```

- Hepatic methylation (arsenic methyltransferase, using S-adenosylmethionine) was once considered detoxification; trivalent methylated intermediates are now known to be genotoxic and to inhibit enzymes more strongly than arsenite.

4.3 Clinical pathology

- Acute poisoning (rare from food): garlic odour, severe cholera-like watery diarrhoea (rice-water stools), vomiting, hypotension, cardiac arrhythmia (QT prolongation), haemolysis, delirium, death; lethal dose of arsenic trioxide about 100 to 300 mg; delayed peripheral neuropathy in survivors; Mees' lines (transverse white bands) on nails 4 to 6 weeks later.
- Chronic arsenicosis (the public health problem): raindrop pigmentation (hyper- and hypopigmented spots on trunk), palmoplantar hyperkeratosis, Bowen's disease, skin cancer (squamous and basal cell), lung and bladder cancer (Group 1 carcinogen), peripheral neuropathy, blackfoot disease (gangrene of toes from peripheral vascular disease, Taiwan), hepatomegaly and portal fibrosis, diabetes, cardiovascular disease, reduced intelligence in children.
- JECFA withdrew the PTWI of 15 micrograms per kg in 2010; the benchmark dose lower limit for 0.5 percent increased lung cancer is 3 micrograms per kg per day, within the range of dietary exposures in affected regions.
- Treatment: dimercaprol or succimer for acute poisoning; for chronic exposure, provision of arsenic-free water (deep aquifers, surface water, arsenic removal plants using iron co-precipitation and adsorption) is the only effective measure.

4.4 Statutory limits for arsenic (mg/kg)

Food category	Limit
Polished rice	0.2 inorganic arsenic (Codex 2014, adopted by FSSAI)
Husked (brown) rice	0.35 inorganic arsenic (Codex 2016)
Packaged drinking water and natural mineral water	0.01 mg/L; Bureau of Indian Standards drinking water acceptable limit 0.01 mg/L (permissible in absence of alternative source 0.05 mg/L)
Edible oils and fats	0.1
Milk; infant foods	0.1 (older general ceiling; harmonised values lower, verify)
Fish and seafood	1.0 (older total arsenic value, reflecting non-toxic arsenobetaine)
Foods not otherwise specified	1.1 (older general ceiling, verify)

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5. Tin (Sn)

5.1 Why tin appears in food

- Tinplate is steel coated with a thin layer of tin, used for cans because tin is corrosion-resistant and non-toxic at low levels. Inorganic tin is poorly absorbed (less than 5 percent) and has low systemic toxicity, but high concentrations irritate the gut.
- Detinning (dissolution of tin into the food) is accelerated by: acidic foods (fruit juices, tomato products, pineapple, rhubarb, pickles; pH below 4), nitrate in the food (from fertilised vegetables and water; nitrate acts as a cathodic depolariser and is the single most important accelerator), oxidising agents, residual oxygen in the headspace, sulphur-containing foods, long storage at high temperature, and opened cans left in the refrigerator (air enters and corrosion accelerates sharply).

```
Acid + nitrate + dissolved oxygen in canned food
        |
        v
Anodic dissolution of the tin coating: Sn -> Sn2+ + 2 electrons
(tin sacrificially protects the underlying steel, which is cathodic)
        |
        v
Stannous ions in the food, rising with storage time and temperature;
hydrogen gas from steel exposure can swell the can ('hydrogen swell',
a non-microbial swell to be distinguished from microbial spoilage)
```

- Control: internal lacquering (epoxy-phenolic, vinyl, polyester enamel) of cans for acidic and nitrate-rich foods, low-nitrate raw materials, exclusion of oxygen (hot filling, vacuum closing), cool storage, and transfer of opened can contents to glass or plastic.

5.2 Toxicity

- Acute gastrointestinal irritation (nausea, vomiting, diarrhoea, abdominal cramps, headache) within 30 minutes to 2 hours when food contains more than about 200 mg/kg tin; recovery is rapid; no chronic effect from inorganic tin at dietary levels; organotins (tributyltin, used as antifouling paint) are far more toxic and are a separate environmental issue.
- JECFA PTWI for inorganic tin: 14 mg per kg body weight per week.

5.3 Statutory ceilings

Food category	Limit
Canned foods (processed and canned products, general)	250 mg/kg (FSSAI, aligned with Codex)

Food category	Limit
Canned beverages, including canned fruit juices and nectars	150 mg/kg (Codex, adopted by FSSAI in the harmonised amendment)
Infant foods (Codex guidance for canned baby foods)	50 mg/kg

[HPSC EXAM TRAP] Tin's statutory ceiling (250 mg/kg) is enormous compared with lead (0.02 to 0.3 mg/kg) because inorganic tin is poorly absorbed and only locally irritant; a pair asserting that tin is 'as toxic as lead' is false. The accelerator to remember for detinning is nitrate, not sugar. A 'hydrogen swell' is chemical, not microbial.

Cross-cutting comparison: biomarkers and antidotes

Metal	Best biomarker	Target organ	Disease name	Antidote or management
Lead	Blood lead; zinc protoporphyrin; ALAD activity	Bone marrow (haem), brain, kidney, peripheral nerve	Plumbism, lead encephalopathy	Calcium disodium edetate, dimercaprol, succimer
Cadmium	Urinary cadmium; urinary beta-2-microglobulin	Kidney proximal tubule, bone, lung	Itai-itai disease	No effective chelator once deposited; stop exposure, calcium and vitamin D
Mercury (methyl)	Hair mercury; blood mercury	Brain (cerebellum, visual cortex), foetus	Minamata disease	Succimer or penicillamine of limited value; stop exposure
Mercury (inorganic)	Urinary mercury	Kidney	Mercurial nephrosis, acrodynia	Dimercaprol, succimer
Arsenic	Urinary total and speciated arsenic; hair and nails	Skin, lung, bladder, nerves, vessels	Arsenicosis, blackfoot disease	Dimercaprol, succimer; safe water
Tin	Faecal tin (poorly absorbed)	Gastrointestinal tract	Acute gastroenteritis	Supportive; none needed

Part 3B: Residues & Process Contaminants

PART 3B: AGROCHEMICAL RESIDUES, VETERINARY DRUGS AND PROCESS-INDUCED CONTAMINANTS

1. Pesticide Classes

1.1 Organochlorines

Compound	Status in India	Note
DDT (dichlorodiphenyltrichloroethane)	Banned for agriculture since 1989; permitted only for public health vector control (malaria, kala-azar)	Metabolites DDE and DDD; half-life in soil 2 to 15 years; eggshell thinning in raptors; Stockholm Convention persistent organic pollutant
BHC or HCH (hexachlorocyclohexane; the gamma isomer is lindane)	Technical HCH banned 1997; lindane banned 2013	Classic contaminant of milk, butter and ghee in Indian surveys
Endosulfan	Banned by Supreme Court order 2011, phased out	Kasaragod (Kerala) aerial spraying tragedy
Aldrin, dieldrin, chlordane, heptachlor, endrin, toxaphene	Banned	All Stockholm Convention persistent organic pollutants

- Properties: highly lipophilic, chemically stable, resistant to biodegradation, hence persistent in soil and water, bioaccumulate in adipose tissue and biomagnify through the food chain; excreted in milk, so they contaminate dairy fat and breast milk.
- Mechanism: DDT holds the voltage-gated sodium channel open (repetitive firing, tremor, convulsions); cyclodienes (aldrin, dieldrin, endosulfan) and lindane antagonise the GABA-A chloride channel (loss of inhibition, seizures). Chronic effects: endocrine disruption (DDE is anti-androgenic), liver enzyme induction, suspected carcinogenicity (DDT Group 2A, lindane Group 1).
- Analysis: fat extraction, clean-up by gel permeation or florisil, gas chromatography with electron capture detector.

1.2 Organophosphates

Compound	Note
Malathion	Low mammalian toxicity because mammalian carboxylesterases hydrolyse it; used for grain storage and public health
Chlorpyrifos	Widely used; neurodevelopmental concerns; European Union withdrew approval 2020; MRLs reduced
Monocrotophos, methyl parathion, phorate, phosphamidon	Highly toxic (Class I); monocrotophos caused the Bihar mid-day meal deaths of 2013 (23 children)
Dimethoate, quinalphos, profenofos, acephate	Common on vegetables and fruits in Indian monitoring

- Mechanism of acetylcholinesterase inhibition:

```
Organophosphate (phosphorothioate, P=S)
      | hepatic cytochrome P450 oxidative desulphuration (bioactivation)
      v
Oxon (P=O), the active inhibitor
      |
      v
Phosphorylates the serine hydroxyl of the esteratic site of acetylcholinesterase
(covalent, initially reversible by oximes)
```

```
    |
    v
'Ageing': loss of an alkyl group from the phosphorylated enzyme within hours to days
(minutes for soman, 24 to 48 hours for most insecticides); the bond becomes
irreversible; recovery needs new enzyme synthesis (weeks)
    |
    v
Acetylcholine accumulates at muscarinic, nicotinic and central synapses
```

- Clinical picture: muscarinic (DUMBELS: diarrhoea, urination, miosis, bronchorrhoea and bronchospasm, emesis, lacrimation, salivation; bradycardia), nicotinic (fasciculations, weakness, respiratory muscle paralysis, tachycardia), central (anxiety, confusion, seizures, coma). Death from respiratory failure. Intermediate syndrome (proximal weakness 1 to 4 days later) and organophosphate-induced delayed polyneuropathy (2 to 3 weeks later, from inhibition of neuropathy target esterase).
- Treatment: decontamination, atropine (muscarinic antagonist, titrated to drying of secretions), pralidoxime (2-PAM; reactivates the enzyme if given before ageing), diazepam for seizures.
- Biomarker: red cell cholinesterase (true acetylcholinesterase) and plasma pseudocholinesterase depression.
- Analysis: QuEChERS extraction (quick, easy, cheap, effective, rugged, safe), gas chromatography with flame photometric or nitrogen-phosphorus detector, liquid chromatography-tandem mass spectrometry.

1.3 Carbamates

- Examples: carbofuran, carbaryl, aldicarb (the most toxic; the 'Temik' poisoning of watermelons, California 1985), methomyl, propoxur, thiodicarb, carbosulfan.
- Mechanism: carbamylate the same serine of acetylcholinesterase, but the carbamyl-enzyme bond hydrolyses spontaneously within minutes to hours; no ageing; so inhibition is reversible and short-lived, symptoms are similar but briefer, central effects are less (poor blood-brain barrier penetration).
- Treatment: atropine; pralidoxime is not generally recommended (and was historically thought to worsen carbaryl toxicity) because the enzyme recovers on its own.
- Environmental fate: not persistent, water-soluble, do not bioaccumulate; carbofuran is highly toxic to birds and was banned in the European Union.

1.4 Synthetic pyrethroids

- Derived from natural pyrethrins of Chrysanthemum cinerariaefolium but with photostable structures: Type I (no alpha-cyano group): permethrin, allethrin, bifenthrin; Type II (alpha-cyano group): cypermethrin, deltamethrin, fenvalerate, lambda-cyhalothrin.
- Mechanism: prolong the open state of voltage-gated sodium channels in insect nerve (Type I: repetitive firing and tremor, T-syndrome; Type II: persistent depolarisation, choreoathetosis and salivation, CS-syndrome); Type II also act on chloride channels.
- Mammalian safety: rapidly hydrolysed by esterases and oxidised by cytochrome P450, poorly absorbed through skin, so acute mammalian toxicity is low (selectivity ratio insect to mammal about 1,000); the main human effect is paraesthesia of exposed skin; highly toxic to fish and bees. Piperonyl butoxide synergist inhibits insect P450.

1.5 Neonicotinoids and others (for completeness)

- Imidacloprid, thiamethoxam, acetamiprid: agonists of insect nicotinic acetylcholine receptors; systemic in plants; implicated in bee decline; low mammalian toxicity.
- Fungicides (mancozeb, carbendazim), herbicides (glyphosate, paraquat; paraquat has no antidote and causes pulmonary fibrosis) and fumigants (aluminium phosphide, phosphine, the commonest fatal poisoning in rural India; methyl bromide, phased out) are other residue classes with MRLs.

1.6 Comparative table of insecticide classes

Property	Organochlorines	Organophosphates	Carbamates	Pyrethroids
Target	Sodium channel (DDT); GABA chloride channel (cyclodienes)	Acetylcholinesterase, irreversible after ageing	Acetylcholinesterase, reversible	Sodium channel kept open
Persistence	Years	Days to weeks	Days	Days (photostable but degradable)
Bioaccumulation	High (fat, milk)	Low	Low	Low
Acute mammalian toxicity	Moderate	High (some extremely high)	Variable (aldicarb extreme, carbaryl low)	Low
Antidote	None specific; diazepam	Atropine plus pralidoxime	Atropine	None needed; vitamin E for paraesthesia
Indian legal status	Mostly banned	Many restricted; some banned (monocrotophos restricted)	Several restricted	Widely permitted

2. Toxicological Parameters: NOAEL, Safety Factor, ADI and MRL

2.1 Definitions

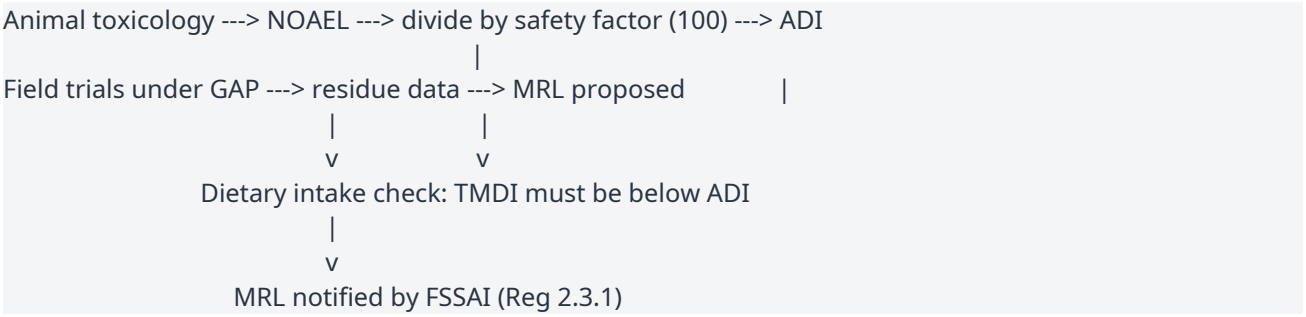
Term	Definition	Who sets it
NOAEL (No Observed Adverse Effect Level)	The highest dose in a chronic animal study (usually the most sensitive species, two-year rat or dog study) at which no statistically or biologically significant adverse effect is observed; expressed in mg per kg body weight per day	Derived from toxicological studies submitted for registration
LOAEL (Lowest Observed Adverse Effect Level)	The lowest dose at which an adverse effect is seen; used with an extra factor when no NOAEL exists	Same
Safety factor (uncertainty factor)	Typically 100: a factor of 10 for extrapolation from animals to humans (interspecies) multiplied by 10 for variation among humans (intraspecies, children, elderly, sick); additional factors of 3 to 10 for severe effects, LOAEL-only data or database deficiencies	Convention of JMPR, JECFA and FSSAI
ADI (Acceptable Daily Intake)	The amount of a chemical in food or drinking water that can be ingested daily over a lifetime without appreciable health risk, expressed in mg per kg body weight per day: ADI = NOAEL / safety factor	JMPR for pesticides, JECFA for additives, contaminants and veterinary drugs; in India, the Registration Committee under the Insecticides Act, 1968, and the FSSAI Scientific Panel
ARfD (Acute Reference Dose)	The amount that can be ingested in one meal or one day without appreciable risk; relevant for acutely toxic pesticides such as organophosphates and carbamates	JMPR
MRL (Maximum Residue Limit)	The maximum concentration of a pesticide residue (mg/kg) legally permitted in or on a food commodity, derived from supervised field trials under Good Agricultural Practice (GAP: approved dose, number of applications, pre-harvest interval); it is a trading standard, not a toxicological threshold, and must be checked so that the Theoretical Maximum Daily Intake does not exceed the ADI	Codex Committee on Pesticide Residues internationally; in India, FSSAI under Regulation 2.3.1 of the Contaminants Regulations, based on recommendations of the Central Insecticides Board and Registration Committee

Term	Definition	Who sets it
PTWI and PTMI	Provisional Tolerable Weekly or Monthly Intake, the equivalent of ADI for cumulative contaminants (lead, cadmium, mercury) that are not deliberately added	JECFA
Tolerance or Default MRL	0.01 mg/kg for any pesticide on any commodity for which no MRL has been fixed (FSSAI, 2018 amendment), functioning as the limit of determination	FSSAI

2.2 Worked derivation

- Suppose the rat two-year NOAEL for a pesticide is 2.5 mg per kg body weight per day. ADI = 2.5 / 100 = 0.025 mg per kg per day. For a 60 kg adult, acceptable intake = 1.5 mg per day.
- Suppose supervised trials under GAP on brinjal give highest residues of 0.4 mg/kg; the MRL is set at 0.5 mg/kg (rounded to the next standard level). If an Indian adult eats 100 g brinjal per day, the intake from brinjal at the MRL is 0.05 mg per day, which is 3.3 percent of the ADI; summing over all commodities on which the pesticide is registered gives the Theoretical Maximum Daily Intake, which must stay below the ADI or the GAP is revised.

2.3 The chain in one diagram



2.4 Legal architecture in India

- Insecticides Act, 1968 and Insecticides Rules, 1971: registration (Central Insecticides Board and Registration Committee), labelling (colour triangle: red for extremely toxic, yellow for highly toxic, blue for moderately toxic, green for slightly toxic), bans and restrictions.
- Food Safety and Standards Act, 2006, Section 19 and Section 20 read with the Contaminants Regulations: no insecticide may be used directly on food except fumigants registered for the purpose; MRLs in Regulation 2.3.1 (Table with over 200 pesticides and commodities); the 0.01 mg/kg default; the Monitoring of Pesticide Residues at National Level scheme (Ministry of Agriculture) generates the surveillance data.
- A few representative MRLs (mg/kg; verify against the current table): DDT in milk and milk products 0.05 on a fat basis (older value 1.25 on fat), in fish 7.0 (older); HCH in milk 0.05 on fat; malathion in cereals 4.0, in vegetables 3.0; chlorpyrifos in cereals 0.05, in vegetables 0.2, in cottonseed 0.05; cypermethrin in vegetables 0.5; carbaryl in vegetables 5.0; carbofuran in vegetables 0.1.

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3. Veterinary Drug Residues and Antibiotics

3.1 Why residues occur

- Antibiotics are used in livestock, poultry, aquaculture and dairy for treatment, prophylaxis and (historically) growth promotion. Residues appear in milk, meat, eggs, honey and shrimp when the withdrawal period (time between the last dose and slaughter or milking) is not observed, when doses are excessive, or when prohibited drugs are used.

3.2 Classes, risks and limits

Class	Examples	Health concern	Typical MRL or status
Beta-lactams	Penicillin G, ampicillin, amoxicillin, cloxacillin, cephalosporins	Hypersensitivity and anaphylaxis in sensitised consumers at microgram doses; inhibition of dairy starter cultures (failed curd, cheese and yoghurt); resistance selection	Codex and FSSAI: benzylpenicillin in milk 4 micrograms per kg (0.004 mg/kg), in muscle 50 micrograms per kg
Tetracyclines	Tetracycline, oxytetracycline, chlortetracycline, doxycycline	Discolouration of developing teeth and bone deposition in children, photosensitivity, resistance	FSSAI (Regulation 2.3.2): tetracycline and oxytetracycline 0.1 mg/kg in fish and shrimp; Codex: 100 micrograms per kg in muscle, 100 in milk, 600 in kidney
Sulphonamides	Sulphamethazine, sulphadiazine, sulphamethoxazole, trimethoprim (combined)	Hypersensitivity, thyroid tumours in rodents (sulphamethazine), resistance	Codex 100 micrograms per kg; FSSAI trimethoprim 0.05 mg/kg in seafood
Aminoglycosides	Streptomycin, gentamicin, neomycin	Nephrotoxicity, ototoxicity	Codex streptomycin 200 micrograms per kg in milk and muscle
Macrolides	Erythromycin, tylosin	Resistance, allergy	Codex 100 to 200 micrograms per kg
Fluoroquinolones	Enrofloxacin, ciprofloxacin	Resistance transfer to human pathogens (Campylobacter, Salmonella); cartilage effects in young animals	Prohibited in aquaculture in India (shrimp export rejections); Codex MRLs for some species
Chloramphenicol	Chloramphenicol	Dose-independent, idiosyncratic aplastic anaemia (irreversible bone marrow failure, often fatal, with no threshold); grey baby syndrome; suspected carcinogen	Zero tolerance: prohibited in all food-producing animals (FSSAI, Codex, European Union, United States); minimum required performance limit of 0.3 micrograms per kg for detection
Nitrofurans	Furazolidone, nitrofurazone, furaltadone, nitrofurantoin (metabolites AOZ, SEM, AMOZ, AHD)	Genotoxic and carcinogenic	Zero tolerance; banned in aquaculture and livestock
Nitroimidazoles	Metronidazole, dimetridazole, ronidazole	Genotoxic, carcinogenic	Prohibited
Hormones and beta-agonists	Diethylstilbestrol, clenbuterol, ractopamine, oestradiol, zeranol	Carcinogenic (DES), cardiac effects (clenbuterol), endocrine disruption	DES and clenbuterol prohibited; India prohibits hormonal growth promoters in food animals
Malachite green (dye)	Used illegally as antifungal in fish farms; leucomalachite green residue	Carcinogenic, genotoxic	Zero tolerance in aquaculture

3.3 Antimicrobial resistance: the mechanism of the public health threat

Sub-therapeutic antibiotic in feed or water
|
v
Selection pressure on gut and environmental bacteria of the animal
|
v
Survival and enrichment of resistant strains (chromosomal mutation or acquisition of plasmid-borne resistance genes, transferred by conjugation)
|
v
Resistant Salmonella, Campylobacter, Escherichia coli and Enterococcus reach

humans through food, water, direct contact and manure on crops
|
v
Treatment failure in human infections; resistance genes transfer to human pathogens (for example, colistin resistance gene mcr-1 from pigs to people)

- Policy response: FSSAI Regulation 2.3.2 (2018) with a table of MRLs for about 103 antibiotics and pharmacologically active substances in meat, milk, eggs and aquaculture, plus an annexed list of substances prohibited for use in any food-producing animal (chloramphenicol, nitrofurans, nitroimidazoles, chlorpromazine, colchicine, dapsone, diethylstilbestrol, fluoroquinolones in aquaculture, glycopeptides, sulphonamides in shrimp, and others); the Coastal Aquaculture Authority list of 20 banned antibiotics for shrimp farming; National Action Plan on Antimicrobial Resistance (2017); Drugs and Cosmetics Rules amendment (2020) requiring withdrawal period on veterinary drug labels.
- Analysis: screening by microbial inhibition tests (Delvotest, Charm) and enzyme-linked immunosorbent assay, confirmation by liquid chromatography-tandem mass spectrometry.

[HPSC EXAM TRAP] Chloramphenicol is not 'restricted to an MRL'; it is prohibited with zero tolerance because aplastic anaemia has no dose threshold. Penicillin residues matter at micrograms per kg because of allergy, not because of general toxicity. A pair claiming that 'cooking destroys all antibiotic residues' is false: tetracyclines and sulphonamides are partly heat-stable, and chloramphenicol survives boiling.

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4. Process-Induced Chemical Contaminants

4.1 Polycyclic Aromatic Hydrocarbons (PAHs)

- Chemistry: fused aromatic ring compounds formed by incomplete combustion and pyrolysis of organic matter above about 350 to 400 degrees Celsius. Benzo[a]pyrene (five rings) is the marker compound; the European Union 'PAH4' group adds benz[a]anthracene, benzo[b]fluoranthene and chrysene. Sixteen PAHs are prioritised by the United States Environmental Protection Agency.
- Formation in food:

Charcoal grilling, barbecuing, tandoor: fat drips onto the fire, pyrolyses, and PAH-laden smoke redeposits on the meat surface; direct flame contact charring; higher with fattier meat, closer flame, longer time
+
Smoking of fish, meat and cheese: wood smoke (hardwood above 400 degrees Celsius) carries PAHs; traditional direct smoking far worse than liquid smoke or indirect smoking with filtered smoke
+
Drying of cereals, oilseeds and cocoa beans by direct combustion gases; roasting of coffee; refined oils (coconut oil from copra dried over fires is the classic high-PAH oil)

- Mechanism of carcinogenesis:

Benzo[a]pyrene (inert procarcinogen)
| cytochrome P450 1A1 and 1B1 (induced by PAHs themselves via the
| aryl hydrocarbon receptor), then epoxide hydrolase, then P450 again
v
Benzo[a]pyrene-7,8-dihydrodiol-9,10-epoxide (BPDE), the ultimate carcinogen
|
v
Covalent bulky adducts at the N2 position of guanine in DNA
|
v
If not repaired by nucleotide excision repair, replication across the adduct

```
causes G to T transversions; the characteristic mutation at codon 157 of the
TP53 tumour suppressor gene in smokers' lung cancer is the molecular signature
    |
    v
Initiation of carcinogenesis (Group 1 human carcinogen: lung, skin, bladder;
dietary association with gastrointestinal cancers)
```

- Limits: European Union: benzo[a]pyrene 2 micrograms per kg and PAH4 12 micrograms per kg in smoked meat and fish (previously 5 micrograms per kg benzo[a]pyrene); 2 micrograms per kg benzo[a]pyrene and 10 PAH4 in oils and fats; 1 microgram per kg benzo[a]pyrene in infant foods. Codex has a code of practice for reducing PAH in smoking and drying. FSSAI has adopted benzo[a]pyrene ceilings in edible oils and smoked products in the harmonised amendments (verify the exact figures).
- Mitigation: indirect smoking, filtered or liquid smoke, lower temperature, remove charred parts, marinate, pre-cook and finish briefly on the grill, keep food away from flames, avoid fat dripping, dry cereals with indirect heat.
- Analysis: saponification or extraction, clean-up on silica or solid-phase extraction, high-performance liquid chromatography with fluorescence detection or gas chromatography-mass spectrometry.

4.2 Acrylamide

- Discovery: Swedish National Food Administration, April 2002, after tunnel workers exposed to acrylamide grout were found to have no higher adduct levels than unexposed people who ate fried foods.
- Formation:

```
Free asparagine (amino acid abundant in potato, wheat, rye, coffee beans)
    +
Reducing sugar carbonyl (glucose, fructose)
    | Maillard reaction, above about 120 degrees Celsius, low moisture,
    | maximal 160 to 180 degrees Celsius
    v
Schiff base -> decarboxylated Amadori product (via Strecker-type chemistry)
    |
    v
Acrylamide (CH2=CH-CO-NH2): the carbon skeleton comes from asparagine,
the sugar carbonyl is the catalyst; minor routes from acrolein and acrylic acid
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- Foods: French fries and potato chips (highest, hundreds to thousands of micrograms per kg), biscuits, crispbread, breakfast cereals, roasted coffee (and coffee substitutes), toasted bread crust, roasted nuts; not formed in boiled or steamed foods (water keeps temperature at 100 degrees Celsius).
- Toxicology: absorbed, distributed widely, metabolised by cytochrome P450 2E1 to glycidamide (an epoxide), which forms DNA adducts (N7-guanine); acrylamide itself binds sulphydryls and causes peripheral neuropathy (occupational), reproductive toxicity in rats; carcinogenic in rodents at multiple sites; Group 2A (probably carcinogenic to humans); haemoglobin adducts are the exposure biomarker.
- Management: no legal maximum limits in most jurisdictions; European Union Regulation 2017/2158 sets benchmark levels (French fries 500 micrograms per kg, potato crisps 750, soft bread 50, breakfast cereals 150 to 300, roasted coffee 400, instant coffee 850) and mandatory mitigation measures; Codex Code of Practice (2009).
- Mitigation: choose low-sugar potato varieties, store potatoes above 6 degrees Celsius (cold storage causes 'cold sweetening' and raises reducing sugars), blanch or soak cut potatoes (removes sugar and asparagine), add asparaginase enzyme to dough, lower frying temperature (below 175 degrees Celsius), fry to golden yellow not brown, add calcium or citric acid, shorter roasting of coffee.
- Analysis: liquid chromatography-tandem mass spectrometry or gas chromatography-mass spectrometry after bromination; isotope dilution with carbon-13 labelled acrylamide.

4.3 Ethyl Carbamate (Urethane)

- Chemistry: NH2-CO-O-C2H5, the ethyl ester of carbamic acid; a natural trace constituent of all fermented foods and beverages; once used as a medicine and as a co-solvent, banned when found carcinogenic in 1943.
- Formation routes:

Route 1 (wine, sake, yoghurt, bread, soy sauce): yeast metabolises arginine (arginase) releasing urea; urea + ethanol -> ethyl carbamate + ammonia, accelerated by heat and storage time; citrulline from lactic acid bacteria (arginine deiminase pathway) is a second precursor via carbamyl phosphate

Route 2 (stone-fruit spirits: kirsch, slivovitz, apricot and plum brandies): cyanogenic glycosides in pits release hydrogen cyanide; cyanide is oxidised (copper and light catalysed) to cyanate; cyanate + ethanol -> ethyl carbamate; this is why distilled stone-fruit spirits have the highest levels

Route 3: diethyl pyrocarbonate (a former wine preservative) + ammonia -> ethyl carbamate; the preservative was banned in 1972 for this reason

- Toxicology: metabolised by cytochrome P450 2E1 to vinyl carbamate and its epoxide, which forms DNA adducts; multisite carcinogen in rodents (lung, liver, lymphoma, mammary); Group 2A (probably carcinogenic to humans); JECFA margin of exposure for heavy consumers of stone-fruit spirits is of concern.
- Typical levels: wines 1 to 30 micrograms per L; sake and sherry higher; stone-fruit brandies up to several mg per L; bread and yoghurt a few micrograms per kg.
- Limits: Canada (the first regulator, 1985): table wine 30 micrograms per L, fortified wine 100, distilled spirits 150, fruit brandies 400, sake 200; United States voluntary targets (wine 15 micrograms per L); Codex Code of Practice (2011) for stone-fruit distillates; European Union target 1 mg per L in stone-fruit spirits.
- Mitigation: low-urea yeast strains, acid urease addition to wine and sake, limiting arginine (vineyard nitrogen management), removing pits before mashing, copper catalysts in stills to remove cyanide, storing spirits in the dark, controlled distillation cuts.
- Analysis: extraction, gas chromatography-mass spectrometry with selected ion monitoring; deuterated internal standard.

4.4 Comparative summary of process contaminants

Contaminant	Precursors	Process condition	Key foods	Ultimate toxic metabolite	Classification	Benchmark or limit
Benzo[a]pyrene (PAH)	Fat, organic matter	Pyrolysis above 350 to 400 degrees Celsius; smoke	Grilled meat, smoked fish, coconut oil, roasted coffee	BPDE	Group 1	EU 2 micrograms per kg in smoked meat and fish
Acrylamide	Asparagine plus reducing sugar	Maillard above 120 degrees Celsius, low moisture	Fries, chips, biscuits, coffee	Glycidamide	Group 2A	EU benchmark 500 micrograms per kg fries
Ethyl carbamate	Urea or citrulline or cyanate plus ethanol	Fermentation, heat, storage, distillation	Wine, sake, stone-fruit spirits, bread	Vinyl carbamate epoxide	Group 2A	Canada 30 micrograms per L wine, 400 fruit brandy
3-MCPD and glycidyl esters (bonus)	Chlorides and partial glycerides	Oil refining deodorisation above 200 degrees Celsius	Refined palm oil, infant formula	3-MCPD, glycidol	2B and 2A	EU 1,250 to 2,500 micrograms per kg in oils
Nitrosamines (bonus)	Nitrite plus secondary amines	Frying of cured meat, direct-fired drying of malt	Bacon, beer (historical), fish sauce	Alkyl diazonium ions	2A	Various national limits

Part 4: Tables, Revision Bank & Model Answers

PART 4: MASTER COMPARATIVE TABLES, REVISION BANK AND MODEL ANSWERS

1. Master Table: Inherent Plant Toxicants and Antinutritional Factors

Toxicant	Chemical class	Principal food	Molecular target	Clinical hallmark	Heat stability	Effective processing	Statutory or safe limit
BOAA / ODAP	Non-protein amino acid (oxalylated diaminopropionic acid)	Khesari dal (Lathyrus sativus)	AMPA glutamate receptor agonist; mitochondrial complex I	Spastic paraparesis of lower limbs (neurolathyrism)	Water-soluble; dry heat ineffective	Hot-water steeping, parboiling, lime soaking (discard water)	Sale prohibited (FSS Prohibition and Restrictions on Sales Regulations, 2011)
Amygdalin, linamarin, dhurrin	Cyanogenic glycosides	Bitter almond and apricot kernels; cassava; sorghum	Cytochrome c oxidase (Fe3+) blocked by cyanide	Histotoxic hypoxia; konzo; goitre	HCN volatile (boiling point 26 degrees Celsius)	Grate, soak, ferment, sun-dry, boil in excess water	Codex: 50 mg/kg sweet cassava, 10 mg/kg flour, 2 mg/kg gari (as HCN)
Vicine, convicine	Pyrimidine glucosides	Faba beans	Divicine and isouramil oxidise glutathione in G6PD-deficient red cells	Acute haemolytic anaemia (favism)	Heat-stable	Avoidance by G6PD-deficient persons; low-vicine cultivars	None; genetic susceptibility
Phytohaemagglutinin	Lectin (glycoprotein)	Raw kidney beans	Brush-border oligosaccharides	Vomiting and diarrhoea within 1 to 3 hours	Destroyed at 100 degrees Celsius; increased by sub-boiling heat	Soak, then boil at least 10 minutes	Cooked beans below 400 hau per g (raw 20,000 to 70,000)
Kunitz and Bowman-Birk inhibitors	Protease inhibitor proteins	Raw soybean, legumes	Trypsin and chymotrypsin; CCK feedback	Pancreatic hypersecretion and hypertrophy, growth depression	Kunitz moist-heat labile; Bowman-Birk more stable	Moist heat 100 degrees Celsius 15 to 20 minutes; autoclaving	Urease test pH change 0.05 to 0.30 for adequately heated soy meal
Phytic acid (IP6)	Inositol hexakisphosphate	Bran, whole grains, legumes	Chelates Zn, Fe, Ca, Mg	Zinc and iron deficiency	Heat-stable	Germination, fermentation, soaking, phytase	Phytate to zinc molar ratio below 15
Solanine, chaconine	Steroidal glycoalkaloids	Green, sprouted potatoes	Acetylcholinesterase; membrane sterols	Gastroenteritis, cholinergic signs after 8 to 12 hours	Stable to boiling and frying	Peel, remove sprouts, store in dark	200 mg/kg total
Progoitrin, sinigrin	Glucosinolates	Cabbage, mustard, rapeseed	Goitrin blocks thyroid peroxidase; thiocyanate blocks iodide uptake	Goitre with marginal iodine	Myrosinase heat-labile; glucosinolate leaches	Blanching, fermentation, iodine supplementation	Rapeseed meal about 30 micromoles per g
Oxalates	Dicarboxylic acid salts	Spinach, rhubarb, colocasia	Chelates calcium; crystallises in kidney	Hypocalcaemia, calcium oxalate stones; raphide	Heat-stable; soluble form leaches	Boil in excess water and discard; add calcium or acid	Lethal oxalic acid 15 to 30 g

Toxicant	Chemical class	Principal food	Molecular target	Clinical hallmark	Heat stability	Effective processing	Statutory or safe limit
				acridity			
Sanguinarine, dihydrosanguinarine	Benzophenanthridine alkaloids	Argemone-adulterated mustard oil	Na+/K+-ATPase; capillary endothelium	Epidemic dropsy, glaucoma, cardiac failure	Stable in oil	None; prevent adulteration	Argemone oil nil (test negative); nitric acid test 0.25 percent; paper chromatography 0.0001 percent

2. Master Table: Marine and Animal Biotoxin Syndromes

Syndrome	Toxin and chemistry	Producer	Vector	Molecular target	Pathognomonic feature	Onset	Action level	Reference analysis
Paralytic shellfish poisoning	Saxitoxin and analogues, water-soluble tetrahydropurine	Alexandrium, Gymnodinium catenatum, Pyrodinium (dinoflagellates)	Bivalves	Sodium channel site 1, block	Ascending paralysis, respiratory arrest, consciousness intact	5 to 30 minutes	80 micrograms STX equivalents per 100 g	Mouse bioassay; HPLC-fluorescence; LC-MS
Diarrhetic shellfish poisoning	Okadaic acid, dinophysistoxins, lipophilic polyethers	Dinophysis, Prorocentrum (dinoflagellates)	Bivalves (hepatopancreas)	Protein phosphatases PP2A and PP1, inhibition	Profuse diarrhoea, no deaths, tumour promoter	30 minutes to 12 hours	160 micrograms OA equivalents per kg	LC-MS; phosphatase assay
Neurotoxic shellfish poisoning	Brevetoxins, lipophilic polyether ladders	Karenia brevis (dinoflagellate)	Bivalves; aerosol	Sodium channel site 5, activation	Temperature reversal, respiratory irritation on beaches	30 minutes to 3 hours	20 mouse units per 100 g (0.8 mg/kg)	Mouse bioassay; ELISA; LC-MS
Amnesic shellfish poisoning	Domoic acid, water-soluble kainoid amino acid	Pseudo-nitzschia (diatom)	Bivalves, anchovies, crabs	Kainate and AMPA glutamate receptors, agonist	Permanent short-term memory loss, hippocampal necrosis	15 minutes to 38 hours	20 mg/kg	HPLC-UV at 242 nm; LC-MS
Ciguatera	Ciguatoxins (polyether), maitotoxin	Gambierdiscus toxicus (benthic dinoflagellate)	Barracuda, grouper, snapper, moray eel	Sodium channel site 5, activation; maitotoxin calcium entry	Cold allodynia or temperature reversal, symptoms for months	1 to 24 hours	0.01 micrograms P-CTX-1 equivalents per kg (United States guidance)	Mouse bioassay; receptor-binding; LC-MS
Scombroid (histamine)	Histamine from histidine by bacterial decarboxylase	Morganella morganii, Klebsiella, Hafnia (bacteria)	Tuna, mackerel, bonito, sardine, mahi-mahi	H1 and H2 receptors (direct)	Flushing, urticaria, pseudo-allergy, cluster of cases	10 minutes to 2 hours	100 mg/kg (Codex hygiene); 200 mg/kg decomposition; 50 mg/kg United States action level	AOAC fluorometric; ELISA; HPLC
Tetrodotoxin poisoning	Tetrodotoxin, aminoperhydroquinazoline	Symbiotic bacteria (Vibrio, Pseudomonas)	Pufferfish ovary, liver, skin	Sodium channel site 1, block	Ascending flaccid paralysis with retained consciousness	10 to 45 minutes	2 mg/kg (10 mouse units per g, Japan)	Mouse bioassay; HPLC post-column fluorescence; LC-MS

3. Master Table: Heavy Metals

Metal	Toxic species	Chief food sources	Enzyme or molecular target	Target organ	Disease or syndrome	Biomarker	JECFA status	Representative FSSAI ceilings (mg/kg)
Lead	Pb2+	Contaminated water, leafy vegetables, spices (lead chromate), glazed ceramics, old canned foods	ALAD (zinc displacement), ferrochelatase, pyrimidine 5'-nucleotidase; calcium mimicry in neurons	Marrow, brain, kidney, peripheral nerve	Plumbism, encephalopathy, microcytic anaemia with basophilic stippling	Blood lead, zinc protoporphyrin	PTWI withdrawn 2010 (no safe level)	Milk and infant formula 0.02; cereals and pulses 0.2; fruits 0.1; fish 0.3; turmeric and tea 10
Cadmium	Cd2+ (metallothionein-bound)	Rice, wheat, leafy vegetables, kidney, liver, crustacean hepatopancreas, oysters, cocoa, tobacco	Sulphydryl enzymes after metallothionein saturation; zinc and calcium displacement	Renal proximal tubule, bone, lung	Itai-itai disease (Jinzu River, Toyama); Fanconi-like tubulopathy	Urinary cadmium, beta-2-microglobulin	PTMI 25 micrograms per kg per month	Rice 0.1 (Codex 0.4); wheat 0.1; leafy vegetables 0.2; fish 0.3; cephalopods 2.0
Mercury	Methylmercury (fish); Hg2+ (salts); Hg0 vapour	Large predatory pelagic fish (shark, swordfish, king mackerel, bigeye tuna)	Sulphydryl groups, tubulin, glutathione; crosses blood-brain barrier and placenta	Cerebellum, visual cortex, foetal brain; kidney (inorganic)	Minamata disease (Chisso, 1956); Hunter-Russell syndrome; congenital Minamata	Hair and blood mercury	PTWI 1.6 micrograms per kg per week (methylmercury)	Fish 0.5; predatory fish 1.0 (methylmercury)
Arsenic	As(III) arsenite most toxic; As(V) arsenate; methylated trivalent forms	Groundwater (Gangetic delta), rice grown in flooded paddies, hijiki	Vicinal dithiols of pyruvate dehydrogenase (As III); phosphate mimicry, arsenolysis (As V)	Skin, lung, bladder, vessels, nerves	Arsenicosis: raindrop pigmentation, keratosis, blackfoot disease, cancers	Urinary speciated arsenic; hair, nails	PTWI withdrawn 2010; BMDL 3 micrograms per kg per day	Polished rice 0.2 inorganic; husked rice 0.35 (Codex); water 0.01 mg/L
Tin	Sn2+ from detinning	Canned acidic and nitrate-rich foods and beverages	Local gastrointestinal irritation only	Gastrointestinal tract	Acute gastroenteritis above about 200 mg/kg	Not needed	PTWI 14 mg per kg per week	Canned foods 250; canned beverages 150

4. Dedicated [HPSC EXAM TRAP] Compendium

Trap 1: Neurolathyrism (BOAA, Lathyrus sativus, spastic paraparesis) versus osteolathyrism (beta-aminopropionitrile, Lathyrus odoratus, skeletal collagen defect).

Trap 2: Linamarin is cassava; amygdalin is bitter almond and apricot kernel; dhurrin is sorghum. Boiling intact bitter cassava is less effective than grating followed by soaking because heat inactivates linamarase before it can free the cyanide.

Trap 3: Favism affects only G6PD-deficient individuals; the glycosides are heat-stable; the mechanism is oxidative haemolysis, not cyanide and not allergy.

Trap 4: Slow cooking of kidney beans at 70 to 80 degrees Celsius increases lectin activity up to five-fold; only boiling destroys it.

Trap 5: Kunitz inhibitor is single-headed (trypsin) and heat-labile; Bowman-Birk is double-headed (trypsin and chymotrypsin) and heat-stable.

Trap 6: Glycoalkaloid limit 200 mg/kg is for the total of solanine plus chaconine; they survive boiling and frying; peeling is the remedy.

Trap 7: Goitrin (from progoitrin) blocks thyroid peroxidase and is not overcome by iodine; thiocyanate (from indole and other glucosinolates, and from cassava) blocks the iodide symporter and is overcome by iodine. Sinigrin gives allyl isothiocyanate, not goitrin.

Trap 8: Argemone: nitric acid test is the screening test (0.25 percent); paper chromatography is the confirmatory, most sensitive test (0.0001 percent); the alkaloid is sanguinarine; the statutory tolerance is nil.

Trap 9: ASP is from a diatom (Pseudo-nitzschia); PSP, DSP and NSP are from dinoflagellates. Saxitoxin and tetrodotoxin block sodium channels; brevetoxin and ciguatoxin activate them; okadaic acid inhibits phosphatases, not channels.

Trap 10: Scombroid poisoning is not an allergy, is not destroyed by cooking or canning, and is prevented by rapid chilling below 4 degrees Celsius after catch.

Trap 11: Lead inhibits ALAD and ferrochelatase (anaemia with normal iron), cadmium causes Itai-itai (Toyama, Japan), methylmercury causes Minamata (Kumamoto, Japan), arsenic causes blackfoot disease (Taiwan) and arsenicosis (West Bengal). Do not cross-match the place names.

Trap 12: ADI = NOAEL / 100 (not multiplied). MRL is a Good Agricultural Practice trading standard, not a toxicological threshold; the default MRL where none exists is 0.01 mg/kg.

Trap 13: Carbamate acetylcholinesterase inhibition is reversible without ageing; pralidoxime is for organophosphates. Chloramphenicol has zero tolerance (aplastic anaemia has no threshold); penicillin's milk MRL is 4 micrograms per kg because of allergy.

Trap 14: Acrylamide needs asparagine plus reducing sugar above 120 degrees Celsius and is absent from boiled food; benzo[a]pyrene comes from pyrolysis above 350 to 400 degrees Celsius and smoke; ethyl carbamate comes from urea or cyanate plus ethanol in fermented and distilled products.

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5. Stage 1 Revision Bank: Count-Based Pair Questions (exactly 4)

Instruction: For each question, three statements are given. Determine how many are correct and choose Only 1 / Only 2 / All 3 / None. Verification uses N = c1 + c2 + c3, where c = 1 for a correct pair and c = 0 for an incorrect pair.

Question 1: Legume toxicants

- (i) BOAA, the neurotoxin of Lathyrus sativus, acts as an agonist at AMPA-type glutamate receptors.
- (ii) Amygdalin is the principal cyanogenic glycoside of cassava (tapioca).
- (iii) The Kunitz inhibitor of soybean inhibits chymotrypsin more strongly than trypsin.

Step-by-step verification:

- (i): BOAA is a glutamate analogue and AMPA receptor agonist causing excitotoxic degeneration of corticospinal tracts. Correct. c1 = 1.
- (ii): The cassava glycoside is linamarin (with minor lotaustralin); amygdalin is in bitter almond and apricot kernels. Incorrect. c2 = 0.
- (iii): Kunitz inhibitor is single-headed and inhibits trypsin primarily; chymotrypsin inhibition is a feature of the double-headed Bowman-Birk inhibitor. Incorrect. c3 = 0.
- N = c1 + c2 + c3 = 1 + 0 + 0 = 1.
- Answer: Only 1.

Question 2: Marine biotoxins

- (i) Saxitoxin causes paralysis by blocking voltage-gated sodium channels at the neuromuscular junction.
- (ii) Domoic acid, the amnesic shellfish poison, is produced by the dinoflagellate Alexandrium.
- (iii) Brevetoxins of Karenia brevis act by activating (opening) voltage-gated sodium channels.

Step-by-step verification:

- (i): Saxitoxin plugs site 1 of the sodium channel; nerve conduction fails; respiratory paralysis follows. Correct. c1 = 1.
- (ii): Domoic acid is produced by the diatom Pseudo-nitzschia, not by a dinoflagellate; Alexandrium produces saxitoxin. Incorrect. c2 = 0.
- (iii): Brevetoxins bind site 5 and keep the channel open, causing persistent depolarisation. Correct. c3 = 1.
- N = c1 + c2 + c3 = 1 + 0 + 1 = 2.
- Answer: Only 2.

Question 3: Heavy metals

- (i) Lead causes anaemia by inhibiting delta-aminolevulinic acid dehydratase and ferrochelatase.
- (ii) Itai-itai disease of the Jinzu River basin in Japan was caused by chronic cadmium exposure through rice.
- (iii) The statutory ceiling for tin in canned foods is 250 mg/kg.

Step-by-step verification:

- (i): Lead displaces zinc from ALAD and inhibits ferrochelatase, blocking haem synthesis; zinc protoporphyrin accumulates. Correct. c1 = 1.
- (ii): Cadmium from the Kamioka mine contaminated irrigation water and rice, causing renal tubular damage and osteomalacia. Correct. c2 = 1.
- (iii): FSSAI, in line with Codex, sets 250 mg/kg for canned foods (150 mg/kg for canned beverages). Correct. c3 = 1.
- N = c1 + c2 + c3 = 1 + 1 + 1 = 3.
- Answer: All 3.

Question 4: Residues and toxicological parameters

- (i) Carbamate insecticides cause irreversible 'ageing' of phosphorylated acetylcholinesterase, so pralidoxime is essential in their treatment.
- (ii) Chloramphenicol has a defined Maximum Residue Limit in milk under the FSSAI Contaminants Regulations.
- (iii) The Acceptable Daily Intake is obtained by multiplying the NOAEL by the safety factor of 100.

Step-by-step verification:

- (i): Carbamates carbamylate the enzyme reversibly; the bond hydrolyses spontaneously; there is no ageing; pralidoxime is not routinely indicated. Incorrect. c1 = 0.
- (ii): Chloramphenicol is on the prohibited (zero-tolerance) list for all food-producing animals because aplastic anaemia has no threshold; no MRL exists. Incorrect. c2 = 0.
- (iii): ADI = NOAEL divided by the safety factor (NOAEL / 100), not multiplied. Incorrect. c3 = 0.
- N = c1 + c2 + c3 = 0 + 0 + 0 = 0.
- Answer: None.

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6. Stage 2 Model Answer (10 Marks)

Question: Detail the biochemical etiology, clinical symptoms, and processing detoxification methods for Neurolathyrism and Epidemic Dropsy.

Introduction (1 mark)

Neurolathyrism and epidemic dropsy are the two classical Indian food toxicoses of plant origin: the first arises from an inherent neurotoxic amino acid in a legume consumed as a famine staple; the second arises from adulteration of mustard oil with the seed oil of a weed. Both are entirely preventable by processing or regulatory control, and both remain statutory concerns under the Food Safety and Standards Act, 2006.

Part A: Neurolathyrism (4.5 marks)

Etiology. Khesari dal (Lathyrus sativus) contains 0.1 to 2.5 percent of beta-N-oxalyl-L-alpha,beta-diaminopropionic acid (BOAA or ODAP), a water-soluble non-protein amino acid in which oxalic acid is amide-linked to the beta-amino group of diaminopropionic acid. BOAA is a structural analogue of glutamate and acts as an agonist at AMPA-type ionotropic glutamate receptors on motor neurons. Sustained receptor activation causes sodium and calcium influx (excitotoxicity), inhibition of mitochondrial complex I, oxidative stress and glutathione depletion, leading to selective degeneration of Betz cells and the long corticospinal axons to the lower limbs. Disease follows consumption of khesari as more than about 30 percent of the diet for 2 to 6 months, mainly in young men doing hard labour, and is aggravated by deficiency of sulphur amino acids and zinc.

Clinical features. Prodromal calf cramps and heaviness of the legs are followed by a spastic gait with scissoring and toe-walking. The disease is staged as no-stick, one-stick, two-stick and crawler stages. Examination shows hypertonia, exaggerated knee and ankle reflexes, clonus and an extensor plantar response, with sensation, sphincters and intellect spared. The paraparesis is permanent; there is no antidote.

Detoxification. Because BOAA is water-soluble, every effective method is a wet method with discarding of the soak water: (1) steeping dehusked dal in hot water at 60 to 70 degrees Celsius for one hour removes 70 to 80 percent; (2) parboiling (soaking, boiling or steaming, discarding water, sun-drying) removes 80 to 90 percent; (3) soaking in 1 to 2 percent lime water (calcium hydroxide) hydrolyses the oxalyl amide bond and removes up to 90 percent; (4) lactic fermentation gives partial removal. Dry roasting and frying are ineffective. Low-toxin cultivars (ODAP below 0.1 percent) are a breeding solution. Statutorily, the sale of kesari gram, kesari dal, kesari flour and their admixtures is prohibited under the Food Safety and Standards (Prohibition and Restrictions on Sales) Regulations, 2011 (continuing Rule 44A of the Prevention of Food Adulteration Rules).

Part B: Epidemic Dropsy (4 marks)

Etiology. Seeds of Argemone mexicana (prickly poppy, satyanashi), a mustard-field weed, resemble mustard seeds and contaminate or are deliberately added to mustard oil. Argemone oil contains the benzophenanthridine alkaloids sanguinarine (about 0.3 percent) and dihydrosanguinarine (about 1 percent), the latter converted to sanguinarine in the body. Sanguinarine inhibits Na+/K+-ATPase in vascular endothelium and cardiac muscle, increases capillary permeability, dilates capillaries, inhibits pyruvate oxidation and glucose-6-phosphatase, and causes oxidative stress. The result is plasma leakage into tissues, pooling of blood and high-output cardiac strain. Oil with about 1 percent argemone oil is sufficient to cause disease.

Clinical features. Whole families fall ill 1 to 3 weeks after consuming the oil. After a gastrointestinal prodrome there is bilateral, pitting, non-inflammatory oedema of the legs with erythema and tenderness, dilated veins and skin nodules; tachycardia, cardiomegaly and congestive heart failure (the main cause of death, about 5 percent); and raised intra-ocular pressure causing glaucoma, with retinal haemorrhages and possible blindness. Anaemia, fever, hepatomegaly and proteinuria also occur. The Delhi outbreak of 1998 (about 3,000 cases and more than 60 deaths) led to the restriction on sale of loose edible oils.

Detection and control. There is no detoxification; the control is prevention of adulteration, supported by analysis: the nitric acid test (5 mL oil shaken with 5 mL concentrated nitric acid gives a red to orange-red colour; sensitivity about 0.25 percent), the ferric chloride test (orange-brown precipitate on heating; about 1 percent), and the confirmatory paper chromatography of the alcoholic extract, in which sanguinarine fluoresces orange-yellow under ultraviolet light and is detectable to 0.0001 percent; thin-layer and high-performance liquid chromatography quantify the alkaloids. The standard for all edible oils requires the argemone test to be negative (nil tolerance), and adulteration causing injury is punishable under Section 59 of the Act.

Conclusion (0.5 mark)

Neurolathyrism is an inherent-toxicant problem solved by wet processing and sale prohibition; epidemic dropsy is an adulteration problem solved by analytical surveillance and packaged-oil regulation. The Food Safety Officer's role is to enforce both: seize khesari admixtures in the dal trade and sample mustard oil for the argemone test.

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18. Stage 2 Model Answer: 15 Marks

Question

Examine the classification, physiological mechanisms and statutory regulatory framework governing heavy metals and agrochemical residues in food.

Model answer

Heavy metals and agrochemical residues are chemical hazards that may enter food through soil, water, industrial activity, agricultural practices, veterinary treatment, processing equipment and environmental deposition. Their risk depends on chemical form, dose, duration, bioavailability and consumer susceptibility.

Classification

Important toxic elements include lead, cadmium, mercury, arsenic and tin.

- Lead is a neurotoxic and haematotoxic metal.
- Cadmium is a cumulative renal and skeletal toxicant.
- Mercury exists as elemental, inorganic and methylmercury forms.
- Arsenic occurs as trivalent, pentavalent and organic compounds; inorganic arsenic is the major food-safety concern.
- Tin may leach from food cans, particularly into acidic foods.

Agrochemical residues include organochlorine pesticides, organophosphates, carbamates and synthetic pyrethroids. Veterinary residues include beta-lactams, tetracyclines, sulfonamides and prohibited substances such as chloramphenicol.

Physiological mechanisms

Lead

Lead inhibits delta-aminolevulinic acid dehydratase and ferrochelatase, thereby impairing haem synthesis and causing anaemia. It also disrupts calcium signalling and neurotransmitter function. Children may suffer impaired cognition, developmental delay and behavioural changes.

Cadmium

Cadmium induces metallothionein and accumulates in proximal renal tubules. Chronic exposure produces tubular proteinuria, reduced renal function, bone demineralisation and osteomalacia. Severe chronic exposure is associated with Itai-Itai disease.

Mercury

Methylmercury is absorbed efficiently from fish, crosses the blood–brain and placental barriers and causes neurological and developmental toxicity. Bioaccumulation and biomagnification make large predatory fish important exposure sources. Minamata disease is the classical example.

Arsenic

Trivalent arsenic binds sulfhydryl groups and inhibits enzymes. Pentavalent arsenic can interfere with phosphate-dependent metabolism. Chronic exposure causes skin lesions, neuropathy, vascular disease and cancer. Contaminated groundwater and rice are important exposure pathways in affected areas.

Tin

Inorganic tin leached from cans may cause gastrointestinal irritation, nausea, vomiting and diarrhoea at high concentrations.

Organophosphate pesticides

Organophosphates phosphorylate acetylcholinesterase and cause accumulation of acetylcholine. Muscarinic, nicotinic and central nervous system symptoms may occur, including salivation, bronchospasm, fasciculations, convulsions and respiratory failure.

Carbamates

Carbamates reversibly inhibit acetylcholinesterase and produce a similar but generally shorter-lived cholinergic syndrome.

Organochlorines

Organochlorines are persistent, lipophilic and bioaccumulative. They may cause neurotoxicity, endocrine effects, reproductive effects and chronic organ toxicity.

Synthetic pyrethroids

Pyrethroids prolong sodium-channel opening and may cause paraesthesia, tremor and seizures in severe exposure.

Veterinary drugs

Residues can cause allergy, direct toxicity, disruption of gut flora, antimicrobial resistance and treatment failure. Chloramphenicol is prohibited because of the risk of aplastic anaemia.

Regulatory framework

The statutory framework is based on the Food Safety and Standards legislation and the applicable regulations relating to contaminants, toxins and residues. Maximum limits for heavy metals are generally commodity-specific. Separate standards may apply to milk, cereals, pulses, vegetables, spices, meat, fish, canned foods and rice. Therefore, a universal lead, cadmium, mercury or arsenic limit should not be quoted without identifying the food category.

Tin has a commonly examined statutory ceiling of 250 milligrams per kilogram in canned food. For arsenic, the distinction between total arsenic and inorganic arsenic is essential. Fish standards may distinguish total mercury from methylmercury or apply species-specific controls.

Pesticide maximum residue limits are individual pesticide–commodity combinations. A residue below the maximum residue limit is legally acceptable for that combination, whereas the acceptable daily intake is a

health-based exposure estimate for the consumer. The no observed adverse effect level is used in risk assessment, and a safety factor accounts for uncertainty and biological variability.

Veterinary drugs are regulated through permitted residue limits, withdrawal periods and prohibitions. Chloramphenicol is treated as a prohibited residue and is not governed by an ordinary positive tolerance level.

Monitoring and enforcement

Food control requires:

1. Risk-based sampling
2. Representative collection
3. Validated extraction
4. Atomic absorption or inductively coupled plasma mass spectrometry for metals
5. Gas chromatography or liquid chromatography coupled with tandem mass spectrometry for pesticide and veterinary residues
6. Screening followed by confirmatory testing
7. Calibration, recovery and uncertainty assessment
8. Comparison with the current notified legal limit
9. Traceability and recall when violations are confirmed

Conclusion

Heavy metals cause toxicity through enzyme inhibition, oxidative stress, renal accumulation, neurotoxicity and interference with mineral or haem metabolism. Agrochemical residues cause acute cholinergic toxicity, chronic neurological or endocrine effects and antimicrobial resistance. Effective regulation requires commodity-specific maximum limits, scientifically derived acceptable daily intakes, withdrawal periods, prohibition of dangerous substances and reliable laboratory surveillance.

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19. Final Rapid Revision Grid

Topic	One-line recall
Khesari dal	ODAP; excitotoxic spastic paraparesis
Cassava	Linamarin; hydrogen cyanide; cytochrome c oxidase inhibition
Faba bean	Vicine and convicine; glucose-6-phosphate dehydrogenase deficiency; haemolysis
Kidney bean	Phytohaemagglutinin; vigorous boiling
Soybean	Kunitz and Bowman–Birk protease inhibitors
Phytate	Mineral chelator; phytase reduces it
Green potato	Solanine and chaconine; heat-stable glycoalkaloids
Brassica	Glucosinolates, goitrin and thiocyanates
Spinach and colocasia	Oxalates; hypocalcaemia and renal stones
Argemone oil	Sanguinarine; epidemic dropsy
Paralytic shellfish poisoning	Saxitoxin; sodium-channel blockade
Diarrhetic shellfish poisoning	Okadaic acid; phosphatase inhibition
Neurotoxic shellfish poisoning	Brevetoxin; sodium-channel activation
Amnesic shellfish poisoning	Domoic acid; hippocampal injury
Ciguatera	Ciguatoxin; temperature reversal

Topic	One-line recall
Scombroid poisoning	Histamine; poor refrigeration; heat-stable
Pufferfish	Tetrodotoxin; sodium-channel blockade
Lead	Inhibits haem synthesis enzymes
Cadmium	Kidney, bone and Itai-Itai disease
Methylmercury	Fish, biomagnification and fetal neurotoxicity
Arsenic	Groundwater, rice and carcinogenicity
Tin	Canned-food leaching; 250 milligrams per kilogram anchor
Organophosphates	Acetylcholinesterase inhibition
Chloramphenicol	Prohibited; aplastic anaemia concern
Benzo[a]pyrene	Charring and DNA adducts
Acrylamide	Asparagine plus reducing sugars at high temperature
Ethyl carbamate	Fermented and alcoholic foods; genotoxic concern